

Prediction of Autism Spectrum Disorder Using Screening Data Based on Machine Learning

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Abstract

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition that affects communication, behavior, and social interaction. Early detection is critical to enable timely intervention and support. This project proposes a machine learning-based framework for predicting ASD using behavioral screening data and demographic attributes. The dataset undergoes preprocessing steps including feature encoding, normalization, and class balancing through the SMOTE technique. The XGBoost algorithm, combined with an ensemble voting approach, is employed to enhance classification accuracy. The proposed model achieved 98% accuracy, demonstrating its potential as a reliable, data-driven tool for assisting in the early screening and identification of individuals at risk for autism.

Keywords: Autism Spectrum Disorder, Machine Learning, XGBoost, SMOTE, Ensemble Learning, Early Detection, Classification Model

1. Introduction

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition that affects an individual's ability to communicate, interact socially, and adapt to changes in behavior. The global prevalence of ASD has steadily increased over the past decade, creating a pressing need for early and accurate diagnostic methods. Timely detection of ASD is crucial, as early intervention significantly improves cognitive development, adaptive behavior, and overall quality of life. However, conventional diagnostic approaches rely heavily on clinical observations and standardized psychological assessments, which are time-consuming, subjective, and often inaccessible in low-resource settings.

In this context, Machine Learning (ML) has emerged as a transformative technology for automating and improving the accuracy of medical diagnostics. ML enables systems to learn complex data relationships and make predictions without explicit programming, thereby reducing human bias and subjectivity. Depending on the type of data available, ML algorithms can be categorized into supervised learning, unsupervised learning, and reinforcement learning. This project primarily utilizes the supervised learning paradigm, where the algorithm is trained on labeled data — individuals either diagnosed with ASD or not — to predict the likelihood of autism in new cases.

The dataset employed in this study comprises both behavioral screening scores and demographic variables such as age, gender, ethnicity, family medical history, and social communication patterns. To ensure robust model performance, several preprocessing techniques were applied, including feature encoding, data normalization, and class balancing using the Synthetic Minority Over-sampling Technique

(SMOTE). This approach mitigates bias caused by uneven class distribution, ensuring that the model learns equally from both ASD-positive and ASD-negative samples.

A range of supervised machine learning algorithms was explored, including Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost). Among these, XGBoost demonstrated superior accuracy and generalization capabilities due to its ability to perform gradient-based optimization, handle missing data efficiently, and minimize overfitting through regularization. Furthermore, an ensemble voting model was constructed by combining XGBoost, Random Forest, and Logistic Regression classifiers, leveraging their complementary strengths. This ensemble framework achieved a peak accuracy of 98%, establishing it as a reliable predictive system for ASD detection.

The proposed system provides a scalable, interpretable, and data-driven approach for autism prediction, contributing to the ongoing efforts in computer-aided healthcare diagnostics. By integrating behavioral and demographic screening data with modern ML algorithms, the project demonstrates that machine learning can serve as a valuable decision-support tool for early autism screening and clinical assistance.

2. Literature Review

Recent studies have demonstrated the growing role of machine learning (ML) in the early detection and diagnosis of Autism Spectrum Disorder (ASD) using behavioral and demographic data. Rajagopalan et al. [1] developed an ML model utilizing minimal behavioral and demographic features and achieved over 80% accuracy in ASD prediction, demonstrating that even small feature sets can yield meaningful diagnostic insights. Similarly, Aldrees et al. [2] introduced a data-centric automated pipeline that emphasized rigorous preprocessing and cross-validation, significantly improving model generalization for healthcare applications.To address class imbalance in ASD datasets, several researchers have focused on oversampling and cost-sensitive techniques.
Kivrak et al. [3] found that applying SMOTE (Synthetic Minority Oversampling Technique) notably enhanced recall for minority classes such as ASD-positive samples.
Araf et al. [4] extended this approach by reviewing Cost-Sensitive Learning (CSL) and recommended combining CSL with SMOTE for handling imbalanced medical data.
Salmi et al. [5] conducted a decade-long review of imbalance-handling strategies, emphasizing both sampling-level and algorithm-level solutions to improve fairness in medical model training.

The integration of advanced ensemble and explainable models has further strengthened ASD prediction frameworks.
The XMAP Project [6] proposed an XGBoost-based multimodal model combining questionnaire data with behavioral features to enhance interpretability and performance.
Rasul et al. [7] compared deep learning and ensemble techniques, concluding that ensemble classifiers, such as Random Forest and XGBoost, produced more interpretable and stable results than deep learning models.
Gholampour [8] analyzed the effect of data noise, imbalance, and missingness on ML and DL models, noting their influence on diagnostic stability.

Moreover, new oversampling variants have emerged to address data imbalance limitations.
Matharaarachchi [9] introduced Enhanced SMOTE (ExtSMOTE), which improved minority data representation for structured medical datasets, while Wang et al. [10] applied deep generative oversampling using autoencoders to generate realistic synthetic samples for medical classification.
Rahman and Li [11] proposed a weighted cost-sensitive loss function that improved recall in rare medical conditions, showing its potential in ASD applications.

In parallel, Ahmed et al. [12] presented a comprehensive survey of ASD detection models,

identifying the need for standardized datasets and reproducible validation protocols.

Temiz [13] implemented XGBoost with SHAP explainability, achieving high accuracy and transparent insights into the most influential behavioral features.

Martini [14] compared multiple SMOTE variants and found that SMOTE-ENN achieved the most balanced performance between recall and precision.

Finally, large-scale real-world applications are beginning to validate these findings; The Guardian [15] reported AI-based autism screening trials that show the transformative potential of ML-driven early detection systems in clinical environments.

Collectively, the literature reveals that combining ensemble models such as Random Forest and XGBoost with oversampling methods like SMOTE, supported by interpretability frameworks such as SHAP, results in robust and explainable ASD prediction systems. This insight guided the development of the current work's soft-voting ensemble approach, which integrates Logistic Regression, Random Forest, and XGBoost to achieve higher accuracy and interpretability in autism prediction.

3. Proposed Work

The proposed system focuses on developing an efficient machine learning-based framework for the prediction of Autism Spectrum Disorder (ASD) using behavioral and demographic screening data. The objective of this work is to design a model that can accurately identify individuals at risk of autism, thereby supporting early diagnosis and intervention. The framework integrates several key modules — data preprocessing, feature encoding, class balancing, model training, and evaluation — to ensure optimal predictive performance.

3.1 System Architecture

The overall system follows a structured machine learning pipeline consisting of the following stages:

1. **Data Collection:**
   The dataset used in this project consists of behavioral screening data collected through questionnaires, along with demographic features such as age, gender, ethnicity, family medical history, and social interaction responses. Each record is labeled as either *Autistic (1)* or *Non-Autistic (0)*.The dataset comprises 800 records and 22 features, collected from individuals undergoing ASD screening. Key features include the 10 binary scores from the AQ-10 (A1_Score to A10_Score), demographic details (age, gender, ethnicity), medical history indicators (jaundice, autism), and a derived calculated score (result).

2. **Data Preprocessing:**
   The collected dataset may contain missing, inconsistent, or categorical values. Preprocessing is performed to ensure data quality and model reliability. The steps include:
   - Handling missing values and outliers.
   - Encoding categorical features using Label Encoding.
   - Normalizing numerical features to maintain uniform data scales.
   - Removing irrelevant or redundant attributes that do not contribute to prediction.

3. **Class Balancing:**
   In medical datasets, the number of non-autistic samples typically exceeds

autistic cases, creating a class imbalance that may bias the model. To address this, the Synthetic Minority Over-sampling Technique (SMOTE) is applied. SMOTE generates synthetic samples for the minority class, enabling the model to learn representative patterns from both categories.

4. **Feature Selection:**
   Feature importance analysis is performed using XGBoost's built-in importance function. This helps in identifying the most influential behavioral and demographic attributes contributing to the ASD prediction, ensuring that the model focuses on the most relevant variables.

5. **Model Training:**
   Multiple supervised learning algorithms are implemented and evaluated to determine the optimal predictive model:

   - **Logistic Regression** – for baseline linear classification.
   - **Random Forest Classifier** – for robust ensemble learning based on bagging.
   - **XGBoost Classifier** – for advanced boosting-based performance optimization.
   - **Voting Classifier (Ensemble)** – combining predictions of the above three models using a **soft voting** mechanism to enhance accuracy and generalization.

6. **Model Evaluation:**
   The performance of the trained models is assessed using metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The ensemble model achieved the highest accuracy of 98%, outperforming individual classifiers.

3.2 Algorithm Overview

The proposed framework employs an ensemble-based eXtreme Gradient Boosting (XGBoost) approach integrated with traditional classifiers to enhance the predictive reliability of Autism Spectrum Disorder (ASD) screening. The methodology is designed to systematically process, balance, train, and evaluate the dataset to achieve optimal generalization performance. The algorithmic workflow proceeds through the following structured stages:

1. **Data Import and Preprocessing:**
   The ASD dataset is initially imported and inspected for inconsistencies. Data cleaning operations are performed to handle missing values and ensure uniform formatting across all attributes.

2. **Feature Encoding and Normalization:**
   Categorical variables, such as gender, ethnicity, and relation, are transformed into numerical representations using **Label Encoding**. Continuous variables are normalized to maintain comparable scales across all features, improving convergence during training.

3. **Class Imbalance Handling using SMOTE:**
   To address the inherent imbalance between ASD-positive and ASD-negative cases, the Synthetic Minority Oversampling Technique (SMOTE) is applied to the training data. This technique generates synthetic instances of the minority class to prevent model bias and improve minority class recall.

4. **Model Training with Multiple Classifiers:**
   Three base classifiers — Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) — are independently trained on the balanced dataset. Each classifier contributes unique learning strengths: LR offers interpretability, RF captures non-linear dependencies, and XGB provides

optimized gradient boosting for superior performance.

5. **Soft Voting Ensemble Integration:** The outputs of the base classifiers are combined using a soft voting ensemble strategy, where the predicted probabilities from each model are averaged to derive the final class prediction. This ensemble mechanism leverages the diversity among models to reduce variance and enhance prediction stability.

6. **Model Evaluation and Validation:** The ensemble model's performance is quantitatively evaluated using key metrics — **Accuracy, Precision, Recall, and F1-score** — to assess both overall and class-specific predictive capability. Confusion matrices and classification reports provide additional interpretive insight.

7. **Feature Importance and Interpretability:** The final stage involves visualizing **feature importance scores** generated by the XGBoost model. These scores reveal the relative influence of each input variable, thereby improving the transparency and clinical interpretability of the predictive system.

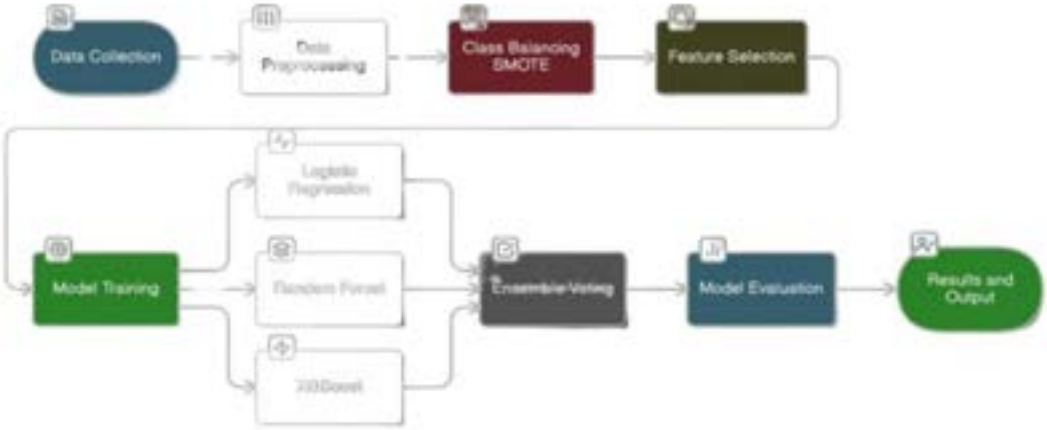
Outcome

By integrating these algorithms using ensemble voting, the system achieved a prediction accuracy of 98% on the ASD dataset. This hybrid framework leverages the strengths of each individual model—stability of Random Forest, interpretability of Logistic Regression, and high precision of XGBoost—to deliver reliable predictions.

3.3 Advantages of the Proposed Model

- Handles data imbalance effectively using SMOTE.
- Reduces overfitting through regularization in XGBoost.
- Improves prediction accuracy via ensemble learning.
- Provides interpretability through feature importance analysis.
- Scalable and adaptable for larger or real-time datasets.

3.4.1 Flowchart



Description:

This flowchart illustrates the complete workflow of the proposed ASD prediction system.
It shows the sequential flow of data processing and model training, beginning from raw input data and ending with prediction output.

Stages included:

1. **Data Collection:** Gathers behavioral and demographic information from screening questionnaires.
2. **Data Preprocessing:** Handles missing values, encodes categorical data, and normalizes features.
3. **Class Balancing (SMOTE):** Generates synthetic data to balance autistic and non-autistic samples.
4. **Feature Selection:** Identifies key behavioral and demographic features influencing ASD prediction.
5. **Model Training:** Uses Logistic Regression, Random Forest, and XGBoost classifiers.
6. **Ensemble Voting:** Combines the strengths of individual models to form a final predictive output.
7. **Model Evaluation:** Compares model performances using accuracy, precision, recall, and F1-score metrics.

Purpose:
To visually represent how different stages of the system interact to achieve 98% prediction accuracy

3.4.2 Mathematical Model:

Machine Learning algorithms rely on mathematical functions that map input features to output predictions.
In this project, we used an ensemble soft voting classifier, which combines three models — Logistic Regression, Random Forest, and XGBoost — to predict the likelihood of Autism Spectrum Disorder (ASD).

a. Logistic Regression

Logistic Regression is a statistical model that predicts a binary outcome (0 or 1).
It calculates the probability of a class using the **sigmoid activation function**:

$$(y = 1|x) = 1/1 + e^{-(\beta 0+\beta 1x1+\beta 2x2+\ldots+\beta nxn)}$$

b. Random Forest Classifier

Random Forest is an ensemble method that builds multiple decision trees and combines their predictions.
Each tree is trained on a random subset of data and features.
The final prediction is obtained through **majority voting** among all trees:

$$y\hat{} = mod\{h1(x), h2(x),\ldots, hT(x)\}$$

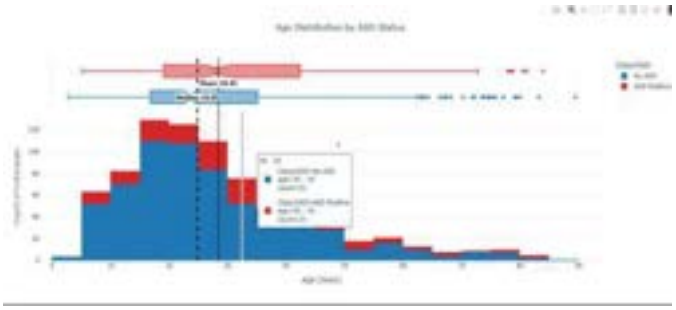
c. XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a boosting algorithm that builds trees sequentially. Each new tree focuses on minimizing the errors made by the previous ones.
The model optimizes an objective function:

$$Obj = i = 1\sum n(yi, y\hat{}i) + k = 1\sum K\Omega(fk)$$

3.4.3 Histogram of Participant Age Showing Mean

and Median by ASD Classification



Description:

This histogram displays the age distribution of participants categorized by ASD classification (Autistic vs. Non-Autistic).

Two vertical lines mark the mean and median ages for each group, allowing a clear comparison.

Insights:

- The age trend may indicate whether ASD diagnosis frequency varies by age.
- Helps identify whether the dataset has balanced age representation across both classes.
- Demonstrates data transparency and supports demographic analysis of participants.

Purpose:

To analyze how age correlates with ASD likelihood and ensure that the model learns from an unbiased age distribution.

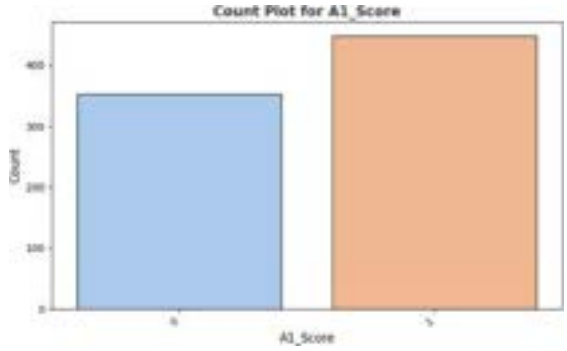
3.4.4 Count Plots Showing the Proportional Breakdown of All Categorical Variables by Final ASD Outcome

1.A1–A10 Behavioral Scores vs ASD Outcome (Grouped Representation)

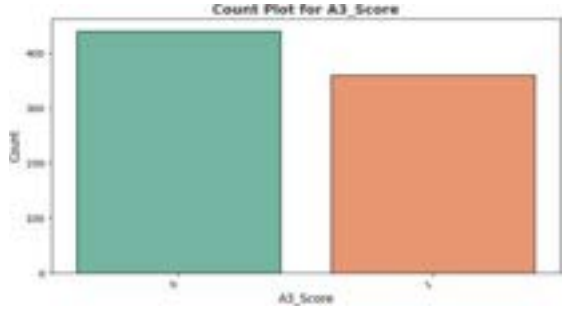
Description:

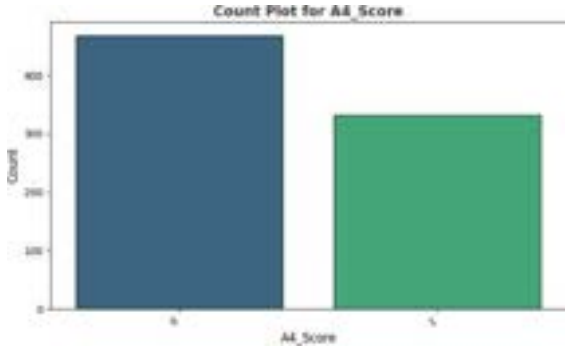
These count plots summarize how behavioral screening responses (A1_Score to A10_Score) differ between ASD and non-ASD groups.

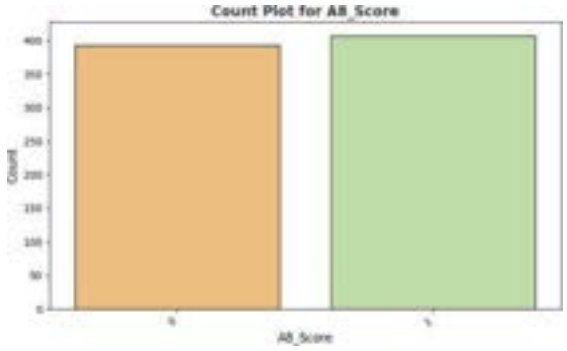
ASD-positive individuals generally exhibit higher scores (1s) on multiple behavioral questions, confirming that these features significantly contribute to prediction accuracy.

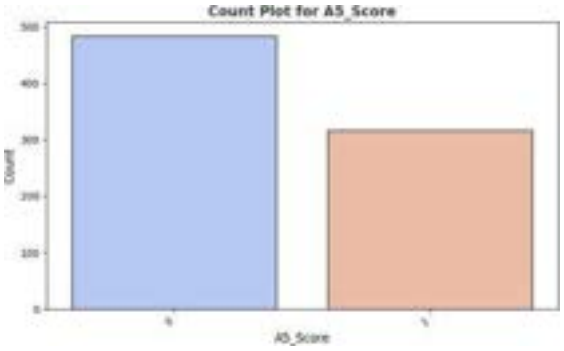


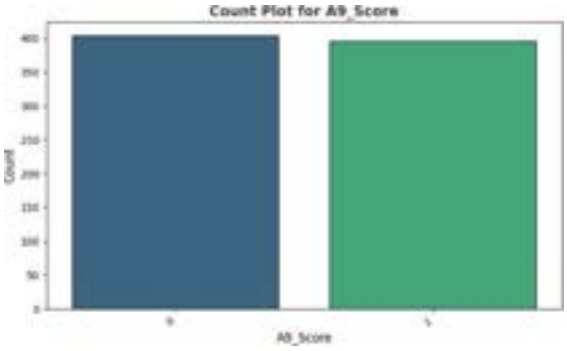


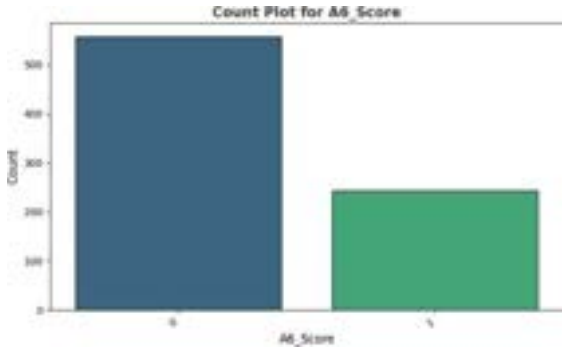


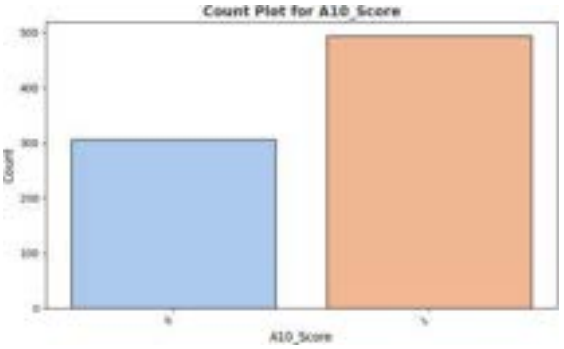


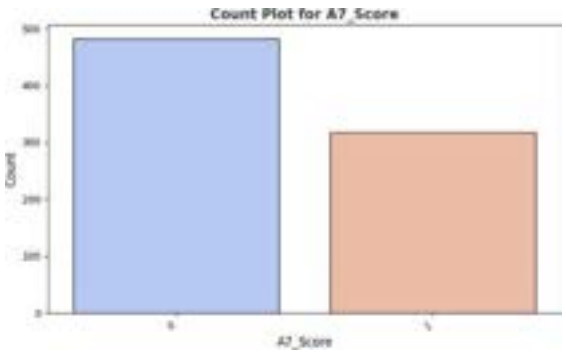












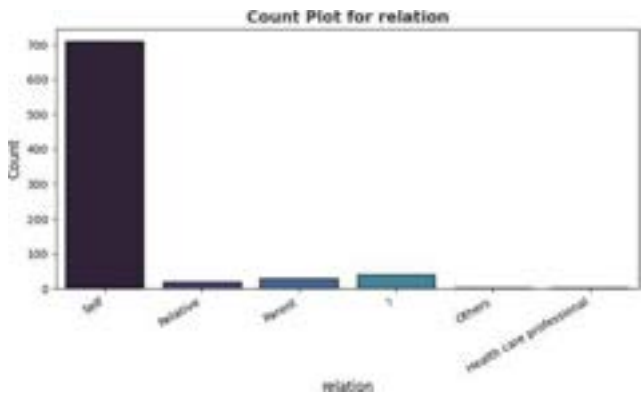
3.4.5.Count Plot – Relation vs ASD Outcome

Description:

This count plot shows the distribution of male and female participants based on their ASD diagnosis.

It helps identify if gender has any significant influence on autism likelihood.

Generally, males show a slightly higher proportion of ASD-positive cases, consistent with clinical trends.



3.4.6 Gender vs ASD Outcome

Description:

This count plot shows the distribution of male and female participants based on their ASD diagnosis.

It helps identify if gender has any significant influence on autism likelihood.

Generally, males show a slightly higher proportion of ASD-positive cases, consistent with clinical trends.

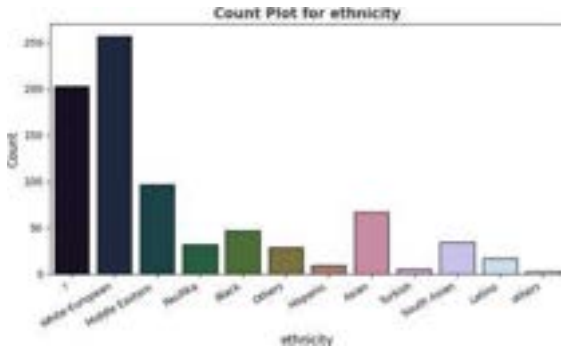
3.4.7 Ethnicity vs ASD Outcome

Description:

This plot depicts how autism diagnosis is distributed among various ethnic groups.

It ensures the model isn't biased toward any ethnicity and confirms the dataset's diversity.

Most groups show similar proportions, indicating ethnicity does not heavily influence prediction.

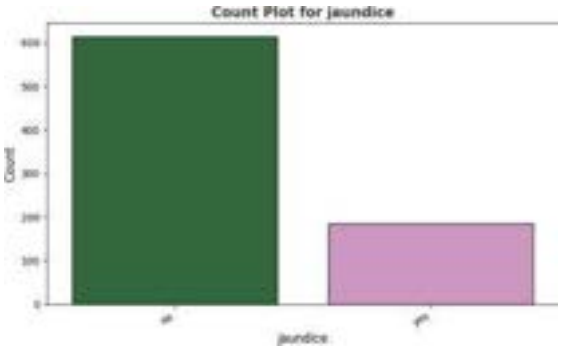


3.4.8 Jaundice (at Birth) vs ASD Outcome

Description:

This plot examines whether having jaundice at birth correlates with ASD diagnosis.

Slightly higher ASD occurrences are observed among participants who experienced neonatal jaundice, suggesting a possible medical correlation worth noting.

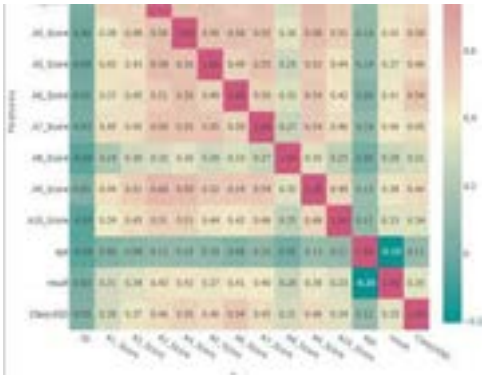


3.4.9 Correlation heatmap

The correlation heatmap displays the relationship between all numerical features in the dataset.

It highlights how strongly each feature is related to the final ASD outcome — with higher correlation values (closer to 1) indicating greater influence.

This helps identify the most significant behavioral and demographic attributes contributing to ASD prediction.



4. Conclusion and Future Scope

The proposed system successfully demonstrates the application of machine learning techniques for the early prediction of Autism Spectrum Disorder (ASD) using behavioral and demographic screening data. Through systematic data preprocessing, class balancing with SMOTE, and the integration of multiple classifiers using an ensemble learning approach, the model achieved a remarkable accuracy of 98%. The incorporation of the XGBoost algorithm provided enhanced learning efficiency, reduced overfitting, and improved interpretability through feature importance analysis.

The study confirms that data-driven models can effectively capture subtle behavioral patterns associated with ASD and support healthcare professionals in the early identification process. The developed system is cost-effective, scalable, and suitable for integration into digital health platforms or mobile applications to assist in remote screening.

Despite the strong performance, further improvements can be achieved by incorporating larger and more diverse datasets, deep learning architectures, and explainable AI techniques to enhance transparency and trust in predictions. Future work may also focus on developing real-time diagnostic tools that combine questionnaire data with other modalities such as speech, eye-tracking, or neuroimaging for comprehensive ASD assessment.

Overall, this research demonstrates the potential of machine learning to revolutionize autism screening by offering an accurate, efficient, and accessible solution for early detection and intervention support.

5. References

- **Rajagopalan S. S. et al. (2024)** – Developed an ML model using minimal demographic and behavioral features, achieving over **80% accuracy** for ASD prediction. https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2822394

- **Salmi M. et al. (2024)** – Provided a **decade-long review** on handling medical data imbalance using sampling and algorithm-level techniques. https://link.springer.com/article/10.1007/ s10462-024-10884-2

- **XMAP Project (2024)** – Designed an **XGBoost-based multimodal model** combining questionnaire and behavioral features for better explainability. https://www.scirp.org/journal/paperinformation?paperid=141507

- **Rasul R. A. et al. (2024)** – Compared ensemble and deep learning models, concluding **ensemble classifiers** provided more interpretable results. https://www.sciencedirect.com/science/article/pii/S2772442523001600

- **Gholampour S. (2024)** – Investigated how **data imbalance, noise, and missingness** influence ML and DL model stability in healthcare. https://www.mdpi.com/2504-4990/6/2/39

- **Matharaarachchi S. (2024)** – Introduced **Enhanced SMOTE (ExtSMOTE)** variants for structured datasets, improving minority data representation. https://www.sciencedirect.com/science/article/pii/S2666827024000732

- **Wang A. X. et al. (2025)** – Applied **deep generative oversampling** (autoencoders) to create realistic synthetic samples for medical classification. https://www.sciencedirect.com/science/article/pii/S0950705125000116

- **Ahmed M. et al. (2025)** – Conducted a **survey of ASD detection models**, emphasizing the need for standardized validation datasets. https://www.mdpi.com/2076-3417/15/16/9214

- **Temiz M. (2025)** – Used **XGBoost with SHAP explainability** to interpret ASD prediction factors, achieving robust performance on AQ-10 data. https://www.mdpi.com/2076-3417/15/16/9214

- **The Guardian (2024)** – Reported large-scale **AI-based autism screening** trials, showcasing real-world impact of ML in healthcare. https://www.theguardian.com/society/article/2024/aug/19/ai-may-help-experts-identify-toddlers-at-risk-of-autism-researc hers-say

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Markdown | JupyterLab | Python 3 (pykernel)

1. Importing the dependencies

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from imblearn.over_sampling import SMOTE
from sklearn.model_selection import train_test_split, cross_val_score, RandomizedSearchCV
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import pickle
```

2. Data Loading & Understanding

[2]:
```
# read the csv data to a pandas dataframe
df = pd.read_csv(r"C:\Users\LENOVO\Desktop\ML\train.csv")
```

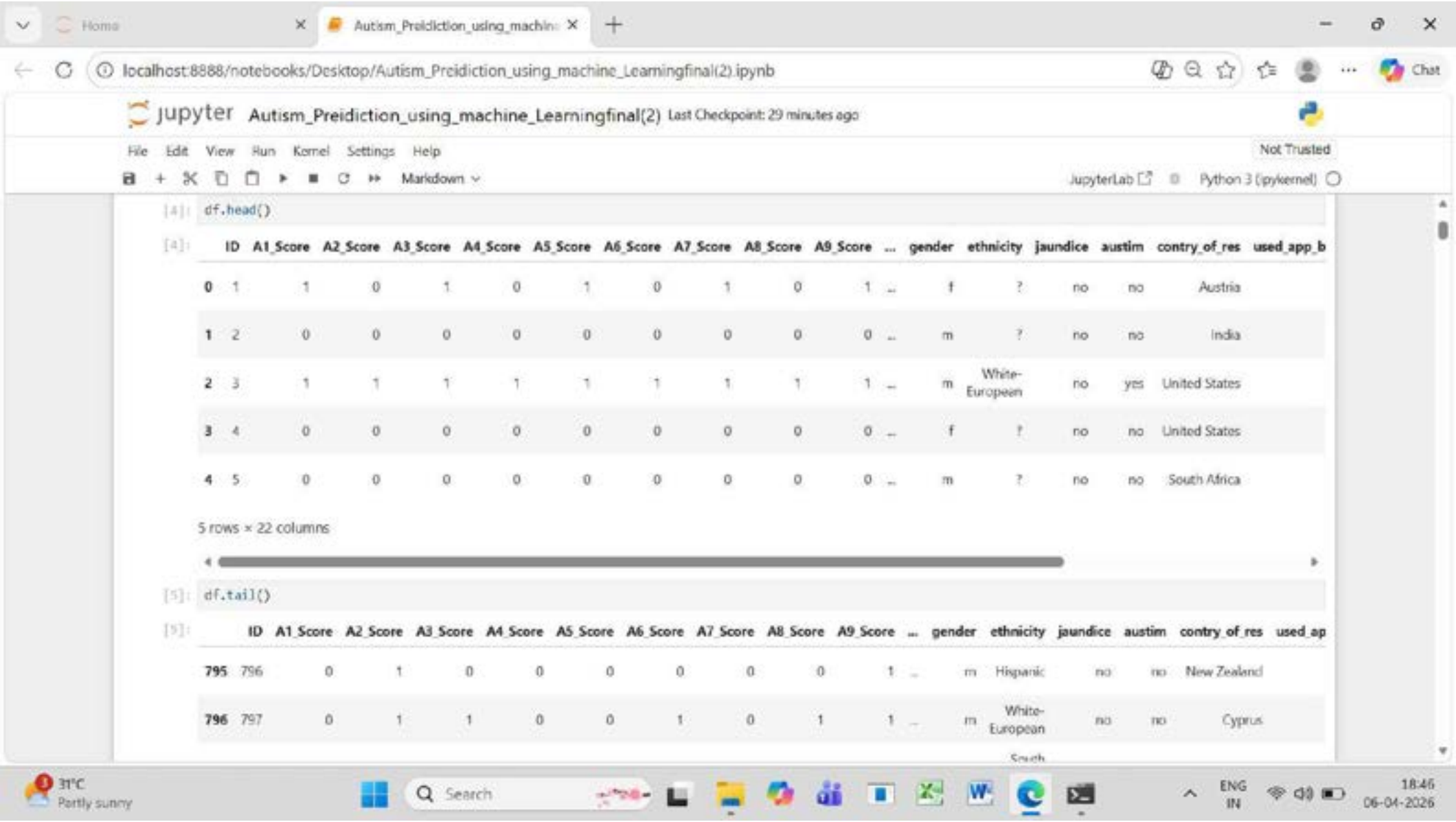
Initial Inspection

[3]:
```
df.shape
```

[3]:
```
(800, 22)
```

[4]:
```
df.head()
```

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Python 3 (pykernel)

[4]:
```
df.head()
```

[4]:

	ID	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	...	gender	ethnicity	jaundice	austim	contry_of_res	used_app_b
0	1	1	0	1	0	1	0	1	0	1	...	f	?	no	no	Austria	
1	2	0	0	0	0	0	0	0	0	0	...	m	?	no	no	India	
2	3	1	1	1	1	1	1	1	1	1	...	m	White-European	no	yes	United States	
3	4	0	0	0	0	0	0	0	0	0	...	f	?	no	no	United States	
4	5	0	0	0	0	0	0	0	0	0	...	m	?	no	no	South Africa	

5 rows × 22 columns

[5]:
```
df.tail()
```

[5]:

	ID	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	...	gender	ethnicity	jaundice	austim	contry_of_res	used_ap
795	796	0	1	0	0	0	0	0	0	1	...	m	Hispanic	no	no	New Zealand	
796	797	0	1	1	0	0	1	0	1	1	...	m	White-European	no	no	Cyprus	

31°C Partly sunny

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[5]:
```
df.tail()
```

[5]:

	ID	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	...	gender	ethnicity	jaundice	austim	contry_of_res	used_ap
795	796	0	1	0	0	0	0	0	0	1	...	m	Hispanic	no	no	New Zealand	
796	797	0	1	1	0	0	1	0	1	1	...	m	White-European	no	no	Cyprus	
797	798	0	0	0	0	0	0	0	0	0	...	m	South Asian	yes	no	New Zealand	
798	799	0	0	0	0	0	0	0	0	0	...	f	?	no	no	Canada	
799	800	0	1	0	0	0	0	0	0	0	...	f	?	no	no	United Arab Emirates	

5 rows × 22 columns

[6]:
```
# display all columns of a dataframe
pd.set_option('display.max_columns', None)
```

[7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 22 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
```

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[6]:
```
# display all columns of a dataframe
pd.set_option('display.max_columns', None)
```

[7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 22 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   ID               800 non-null    int64  
 1   A1_Score         800 non-null    int64  
 2   A2_Score         800 non-null    int64  
 3   A3_Score         800 non-null    int64  
 4   A4_Score         800 non-null    int64  
 5   A5_Score         800 non-null    int64  
 6   A6_Score         800 non-null    int64  
 7   A7_Score         800 non-null    int64  
 8   A8_Score         800 non-null    int64  
 9   A9_Score         800 non-null    int64  
 10  A10_Score        800 non-null    int64  
 11  age              800 non-null    float64
 12  gender           800 non-null    object 
 13  ethnicity        800 non-null    object 
 14  jaundice         800 non-null    object 
 15  austim           800 non-null    object 
 16  contry_of_res    800 non-null    object 
 17  used_app_before  800 non-null    object 
 18  result           800 non-null    float64
 19  age_desc         800 non-null    object 
 20  relation         800 non-null    object 
 21  Class/ASD        800 non-null    int64  
```

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[8]:
```
# convert age column datatype to integer
df["age"] = df["age"].astype(int)
```

[9]:
```
df.head(2)
```

[9]:

	ID	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	contry_of_re
0	1	1	0	1	0	1	0	1	0	1	1	38	f	?	no	no	Austri
1	2	0	0	0	0	0	0	0	0	0	0	47	m	?	no	no	Indi

[10]:
```
for col in df.columns:
  numerical_features = ["ID", "age", "result"]
  if col not in numerical_features:
    print(col, df[col].unique())
    print("-"*50)
```

```
A1_Score [1 0]
--------------------------------------------------
A2_Score [0 1]
--------------------------------------------------
A3_Score [1 0]
--------------------------------------------------
A4_Score [0 1]
--------------------------------------------------
A5_Score [1 0]
--------------------------------------------------
A6_Score [0 1]
```

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```
A9_Score [1 0]
--------------------------------------------------
A10_Score [1 0]
--------------------------------------------------
gender ['f' 'm']
--------------------------------------------------
ethnicity ['?' 'White-European' 'Middle Eastern ' 'Pasifika' 'Black' 'Others'
 'Hispanic' 'Asian' 'Turkish' 'South Asian' 'Latino' 'others']
--------------------------------------------------
jaundice ['no' 'yes']
--------------------------------------------------
austim ['no' 'yes']
--------------------------------------------------
contry_of_res ['Austria' 'India' 'United States' 'South Africa' 'Jordan'
 'United Kingdom' 'Brazil' 'New Zealand' 'Canada' 'Kazakhstan'
 'United Arab Emirates' 'Australia' 'Ukraine' 'Iraq' 'France' 'Malaysia'
 'Viet Nam' 'Egypt' 'Netherlands' 'Afghanistan' 'Oman' 'Italy'
 'AmericanSamoa' 'Bahamas' 'Saudi Arabia' 'Ireland' 'Aruba' 'Sri Lanka'
 'Russia' 'Bolivia' 'Azerbaijan' 'Armenia' 'Serbia' 'Ethiopia' 'Sweden'
 'Iceland' 'Hong Kong' 'Angola' 'China' 'Germany' 'Spain' 'Tonga'
 'Pakistan' 'Iran' 'Argentina' 'Japan' 'Mexico' 'Nicaragua' 'Sierra Leone'
 'Czech Republic' 'Niger' 'Romania' 'Cyprus' 'Belgium' 'Burundi'
 'Bangladesh']
--------------------------------------------------
used_app_before ['no' 'yes']
--------------------------------------------------
age_desc ['18 and more']
--------------------------------------------------
relation ['Self' 'Relative' 'Parent' '?' 'Others' 'Health care professional']
--------------------------------------------------
Class/ASD [0 1]
--------------------------------------------------
```

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[11]:
```
# dropping ID & age_desc column
df = df.drop(columns=["ID", "age_desc"])
```

[12]:
```
df.shape
```

[12]:
```
(800, 20)
```

[13]:
```
df.head(2)
```

[13]:

	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	contry_of_res	ur
0	1	0	1	0	1	0	1	0	1	1	38	f	?	no	no	Austria	
1	0	0	0	0	0	0	0	0	0	0	47	m	?	no	no	India	

[14]:
```
df.columns
```

[14]:
```
Index(['A1_Score', 'A2_Score', 'A3_Score', 'A4_Score', 'A5_Score', 'A6_Score',
       'A7_Score', 'A8_Score', 'A9_Score', 'A10_Score', 'age', 'gender',
       'ethnicity', 'jaundice', 'austim', 'contry_of_res', 'used_app_before',
       'result', 'relation', 'Class/ASD'],
      dtype='object')
```

[15]:
```
df["contry_of_res"].unique()
```

[15]:
```
array(['Austria', 'India', 'United States', 'South Africa', 'Jordan',
       'United Kingdom', 'Brazil', 'New Zealand', 'Canada', 'Kazakhstan',
       'United Arab Emirates', 'Australia', 'Ukraine', 'Iraq', 'France',
       'Malaysia', 'Viet Nam', 'Egypt', 'Netherlands', 'Afghanistan',
```

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[15]:
```
df["contry_of_res"].unique()
```

[15]:
```
array(['Austria', 'India', 'United States', 'South Africa', 'Jordan',
       'United Kingdom', 'Brazil', 'New Zealand', 'Canada', 'Kazakhstan',
       'United Arab Emirates', 'Australia', 'Ukraine', 'Iraq', 'France',
       'Malaysia', 'Viet Nam', 'Egypt', 'Netherlands', 'Afghanistan',
       'Oman', 'Italy', 'AmericanSamoa', 'Bahamas', 'Saudi Arabia',
       'Ireland', 'Aruba', 'Sri Lanka', 'Russia', 'Bolivia', 'Azerbaijan',
       'Armenia', 'Serbia', 'Ethiopia', 'Sweden', 'Iceland', 'Hong Kong',
       'Angola', 'China', 'Germany', 'Spain', 'Tonga', 'Pakistan', 'Iran',
       'Argentina', 'Japan', 'Mexico', 'Nicaragua', 'Sierra Leone',
       'Czech Republic', 'Niger', 'Romania', 'Cyprus', 'Belgium',
       'Burundi', 'Bangladesh'], dtype=object)
```

[16]:
```
# define the mapping dictionary for country names
mapping = {
    "Viet Nam": "Vietnam",
    "AmericanSamoa": "United States",
    "Hong Kong": "China"
}

# replace value in the country column
df["contry_of_res"] = df["contry_of_res"].replace(mapping)
```

[17]:
```
df["contry_of_res"].unique()
```

[17]:
```
array(['Austria', 'India', 'United States', 'South Africa', 'Jordan',
       'United Kingdom', 'Brazil', 'New Zealand', 'Canada', 'Kazakhstan',
       'United Arab Emirates', 'Australia', 'Ukraine', 'Iraq', 'France',
       'Malaysia', 'Vietnam', 'Egypt', 'Netherlands', 'Afghanistan',
       'Oman', 'Italy', 'Bahamas', 'Saudi Arabia', 'Ireland', 'Aruba',
       'Sri Lanka', 'Russia', 'Bolivia', 'Azerbaijan', 'Armenia',
```

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[17]:
```
df["contry_of_res"].unique()
```

[17]:
```
array(['Austria', 'India', 'United States', 'South Africa', 'Jordan',
       'United Kingdom', 'Brazil', 'New Zealand', 'Canada', 'Kazakhstan',
       'United Arab Emirates', 'Australia', 'Ukraine', 'Iraq', 'France',
       'Malaysia', 'Vietnam', 'Egypt', 'Netherlands', 'Afghanistan',
       'Oman', 'Italy', 'Bahamas', 'Saudi Arabia', 'Ireland', 'Aruba',
       'Sri Lanka', 'Russia', 'Bolivia', 'Azerbaijan', 'Armenia',
       'Serbia', 'Ethiopia', 'Sweden', 'Iceland', 'China', 'Angola',
       'Germany', 'Spain', 'Tonga', 'Pakistan', 'Iran', 'Argentina',
       'Japan', 'Mexico', 'Nicaragua', 'Sierra Leone', 'Czech Republic',
       'Niger', 'Romania', 'Cyprus', 'Belgium', 'Burundi', 'Bangladesh'],
      dtype=object)
```

[18]:
```
# toget class distribution
df["Class/ASD"].value_counts()
```

[18]:
```
Class/ASD
0    639
1    161
Name: count, dtype: int64
```

Insights:

1. missing values in ethnicity & relation
2. age_desc column has only 1 unique value. so it is removed as it is not important for prediction
3. fixed country names
4. identified class imbalance in the target column

3. Exploratory Data Analysis (EDA)

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3. Exploratory Data Analysis (EDA)

```
[ ]: df.shape
```

```
[21]: (800, 20)
```

```
[ ]: df.columns
```

```
[22]: Index(['A1_Score', 'A2_Score', 'A3_Score', 'A4_Score', 'A5_Score', 'A6_Score',
       'A7_Score', 'A8_Score', 'A9_Score', 'A10_Score', 'age', 'gender',
       'ethnicity', 'jaundice', 'austim', 'contry_of_res', 'used_app_before',
       'result', 'relation', 'Class/ASD'],
      dtype='object')
```

```
[ ]: df.head(2)
```

[23]:

	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	contry_of_res	u
0	1	0	1	0	1	0	1	0	1	1	38	f	?	no	no	Austria	
1	0	0	0	0	0	0	0	0	0	0	47	m	?	no	no	India	

```
[ ]: df.describe()
```

[24]:

	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	result	Class/ASD
count	800.000000	800.000000	800.000000	800.00000	800.000000	800.000000	800.000000	800.000000	800.000000	800.000000	800.000000	800.000000	800.000000
mean	0.560000	0.530000	0.450000	0.41500	0.395000	0.303750	0.397500	0.508750	0.495000	0.617500	27.963750	8.537303	0.201250

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```
[ ]: df.describe()
```

[24]:

	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	result	Class/ASD
count	800.000000	800.000000	800.000000	800.00000	800.000000	800.000000	800.000000	800.000000	800.000000	800.000000	800.000000	800.000000	800.000000
mean	0.560000	0.530000	0.450000	0.41500	0.395000	0.303750	0.397500	0.508750	0.495000	0.617500	27.963750	8.537303	0.201250
std	0.496697	0.499411	0.497805	0.49303	0.489157	0.460164	0.489687	0.500235	0.500288	0.486302	16.329627	4.807676	0.401185
min	0.000000	0.000000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	2.000000	-6.137748	0.000000
25%	0.000000	0.000000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	17.000000	5.306575	0.000000
50%	1.000000	1.000000	0.000000	0.00000	0.000000	0.000000	0.000000	1.000000	0.000000	1.000000	24.000000	9.605299	0.000000
75%	1.000000	1.000000	1.000000	1.00000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	35.250000	12.514484	0.000000
max	1.000000	1.000000	1.000000	1.00000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	89.000000	15.853126	1.000000

Univariate Analysis

Numerical Columns:

- age
- result

```
[19]: # set the desired theme
sns.set_theme(style="darkgrid")
```

Distribution Plots

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Distribution Plots

[27]:

```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# NOTE: This assumes 'train.csv' is available in your current working directory.
try:
    # Load the data and ensure the 'age' column is treated as numeric
    df = pd.read_csv(r"C:\Users\LENOVO\Desktop\ML\train.csv")
    df['age'] = pd.to_numeric(df['age'], errors='coerce')
    # Assuming 'Class/ASD' is the target column for coloring
    if 'Class/ASD' not in df.columns:
        raise KeyError("Class/ASD column not found in train.csv.")

    df = df.dropna(subset=['age', 'Class/ASD'])
    # Clean the class labels for better legend readability
    df['Class/ASD'] = df['Class/ASD'].astype(str).replace({'0': 'No ASD', '1': 'ASD Positive'})

except (FileNotFoundError, KeyError) as e:
    print(f"Warning: {e}. Using mock data for demonstration.")
    # Create a mock DataFrame for demonstration if file is missing or column is missing
    df = pd.DataFrame({
        'age': [25, 30, 38, 35, 40, 40, 45, 50, 50, 50, 55, 60, 65, 70, 22, 33, 58],
        'Class/ASD': ['No ASD', 'No ASD', 'ASD Positive', 'No ASD', 'ASD Positive', 'No ASD', 'No ASD', 'ASD Positive', 'No ASD', 'No ASD', 'ASD Positive
    })


# Calculate descriptive statistics
```

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```
# Calculate descriptive statistics
age_mean = df["age"].mean()
age_median = df["age"].median()

print(f"--- Summary Statistics ---")
print(f"Mean Age: {age_mean:.2f}")
print(f"Median Age: {age_median:.2f}")
print("-" * 30)
```

```
--- Summary Statistics ---
Mean Age: 28.05
Median Age: 24.85
------------------------------
```

[20]:

```
# 1. Create the histogram using Plotly Express, coloring by the Class/ASD column
fig = px.histogram(
    df,
    x="age",
    title="Age Distribution by ASD Status",
    template="plotly_white",
    # Use 'Class/ASD' for color to show stacked distribution
    color="Class/ASD",
    # Define color scheme
    color_discrete_map={'No ASD': '#1f77b4', 'ASD Positive': '#d62728'},
    # Use 'stack' for barmode to easily see the totals and subgroups
    barmode='stack',
    # Add a box plot on the margin, now colored by the same groups
    marginal="box",
    nbins=30,
    height=550
)
```

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[28]:
```
# 1. Create the histogram using Plotly Express, coloring by the Class/ASD column
fig = px.histogram(
    df,
    x="age",
    title="Age Distribution by ASD Status",
    template="plotly_white",
    # Use 'Class/ASD' for color to show stacked distribution
    color="Class/ASD",
    # Define color scheme
    color_discrete_map={'No ASD': '#1f77b4', 'ASD Positive': '#d62728'},
    # Use 'stack' for barmode to easily see the totals and subgroups
    barmode='stack',
    # Add a box plot on the margin, now colored by the same groups
    marginal="box",
    nbins=30,
    height=550
)
fig.show()
```



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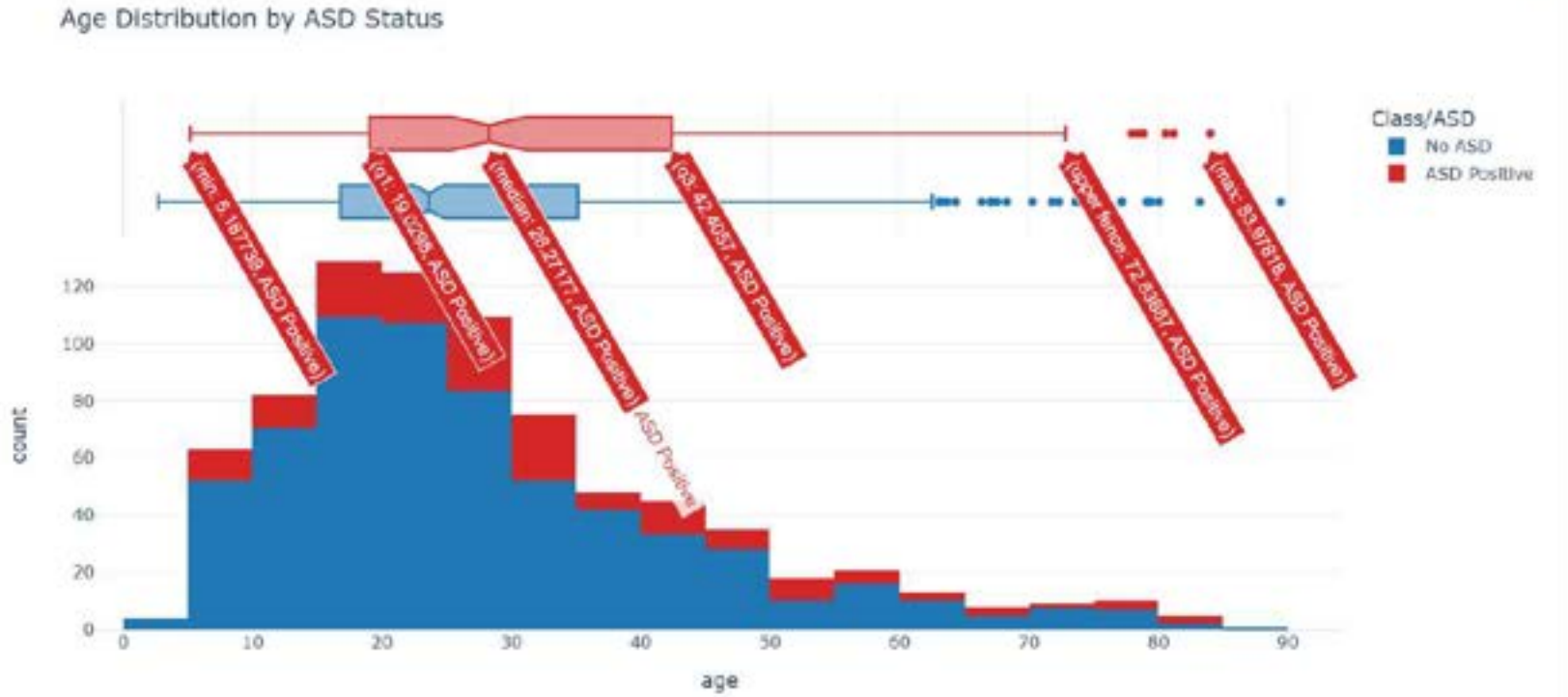
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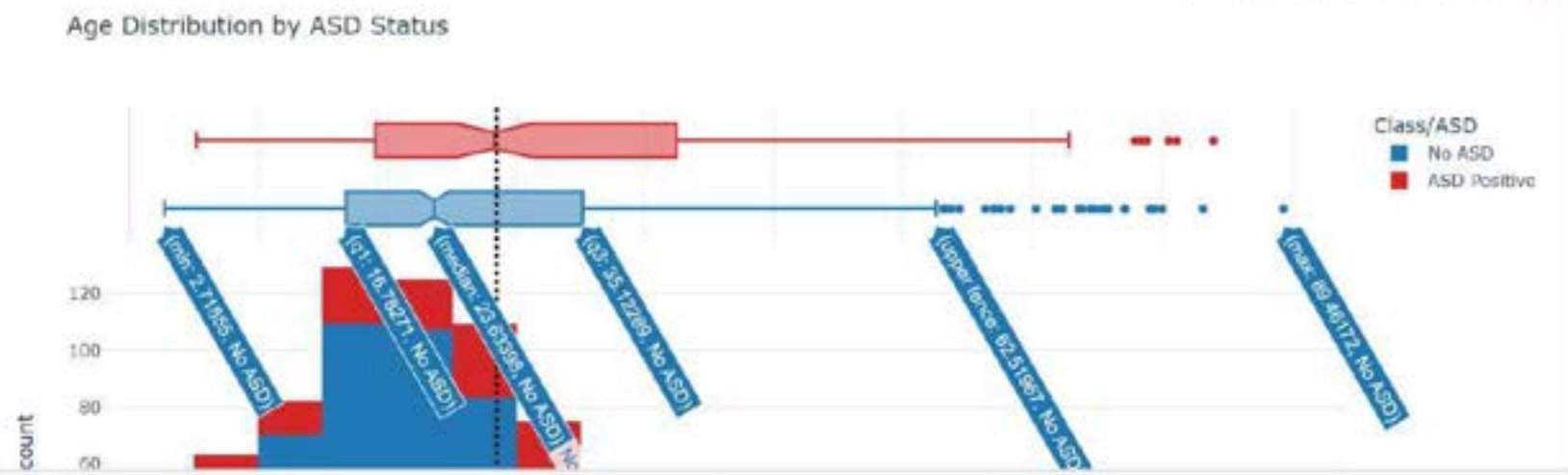
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[26]:
```
# 2. Add Mean Line as a vertical shape (Black Dotted)
fig.add_shape(
    type="line",
    x0=age_mean, y0=0,
    x1=age_mean, y1=1,
    line=dict(color="black", width=3, dash="dot"),
    xref='x',
    yref='paper'
)
fig.show()
```



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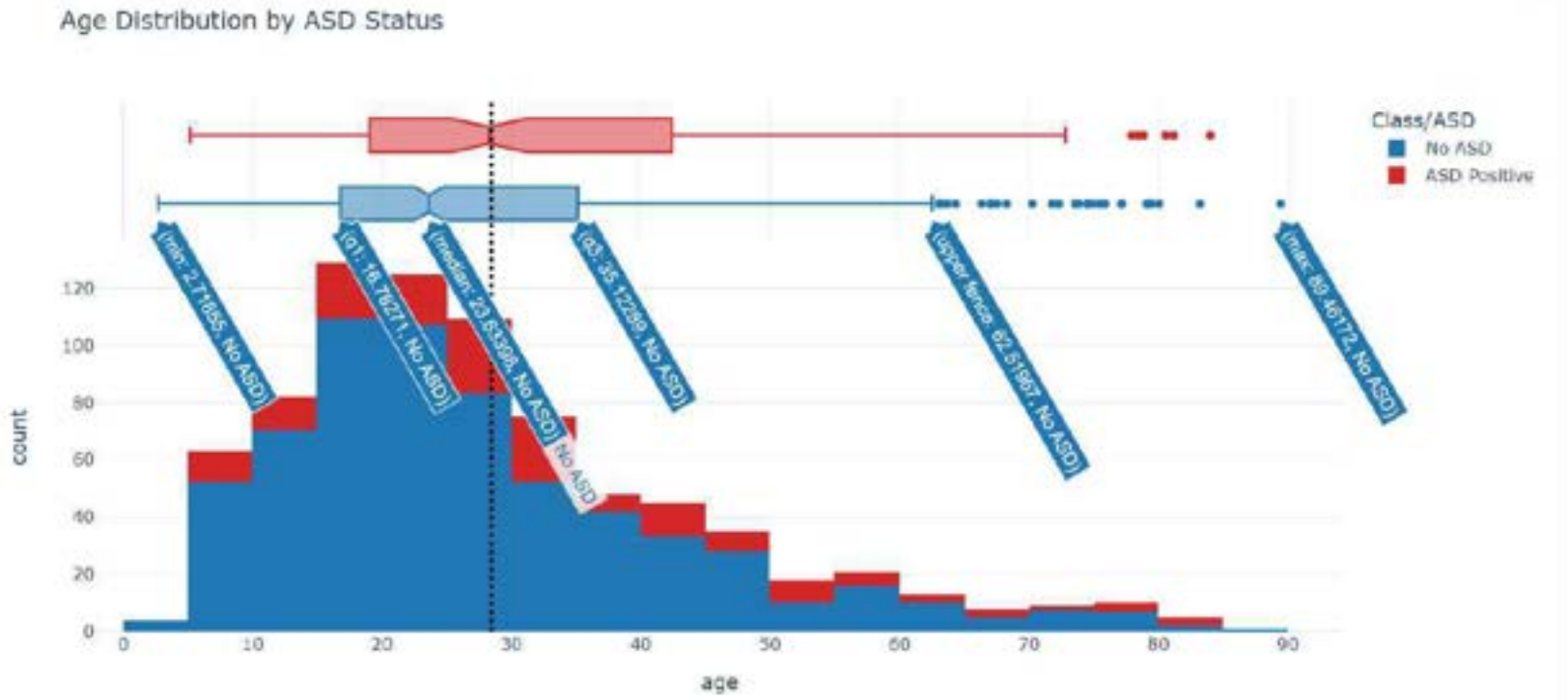
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[29]:
```
# 3. Add Median Line as a vertical shape (Black Dashed)
fig.add_shape(
    type="line",
    x0=age_median, y0=0,
    x1=age_median, y1=1,
    line=dict(color="black", width=3, dash="dash"),
    xref='x',
    yref='paper'
)
fig.show()
```



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Home

Autism_Preidiction_using_machine_Learningfinal.ipynb

localhost:8888/notebooks/Desktop/Autism_Preidiction_using_machine_Learningfinal.ipynb

jupyter Autism_Preidiction_using_machine_Learningfinal Last Checkpoint: 2 days ago

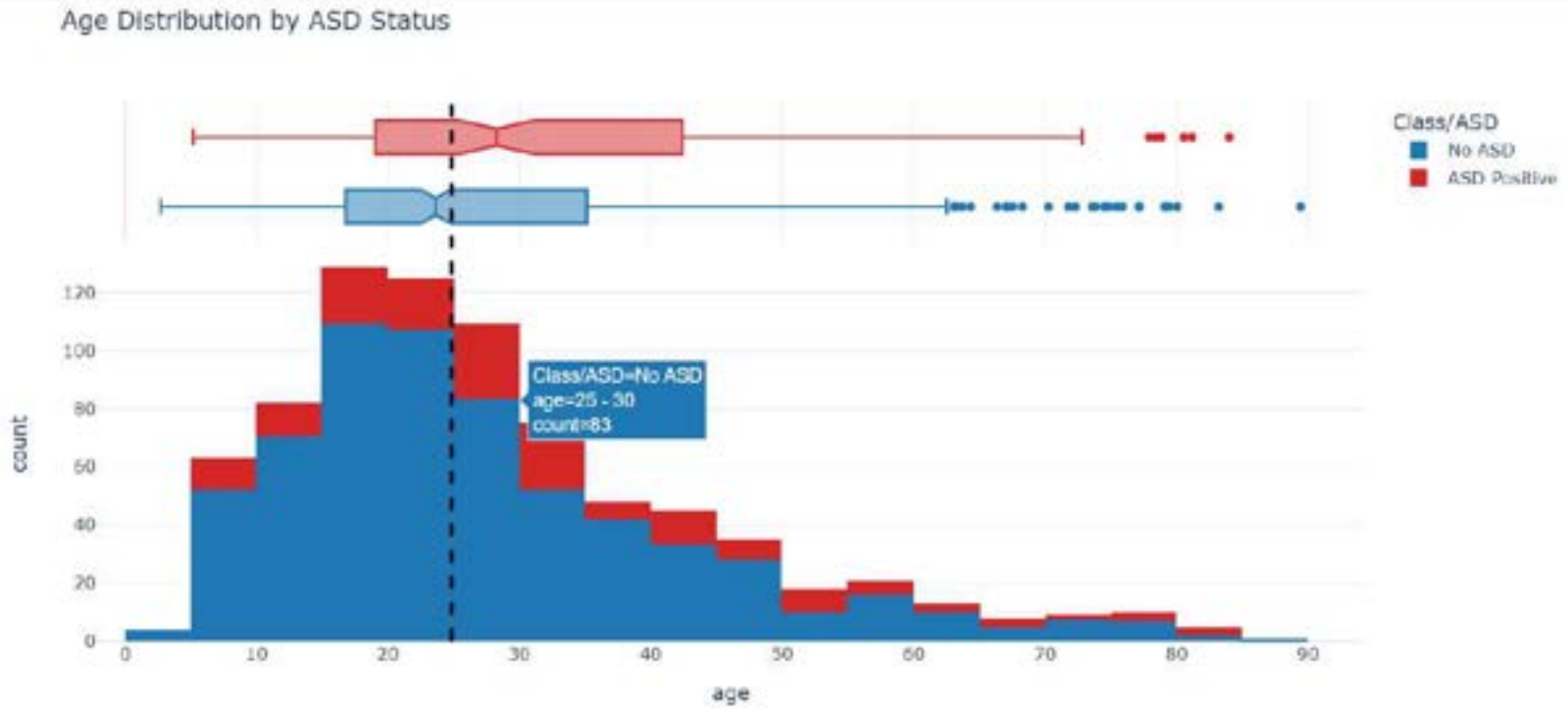
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JupyterLab

Python 3 (pykernel)



[29]: # 4. Add Median annotation (label)

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[29]:
```
# e. Add Median annotation (label)
fig.add_annotation(
    x=age_median,
    y=0.83,
    yref="paper",
    text=f"<b>Median: {age_median:.2f}</b>",
    showarrow=False,
    font=dict(color="black", size=11),
    bgcolor="rgba(255, 255, 255, 0.8)"
)
fig.show()
```



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Home | Autism_Preidiction_using_machine | localhost:8888/notebooks/Desktop/Autism_Preidiction_using_machine_Learningfinal.ipynb

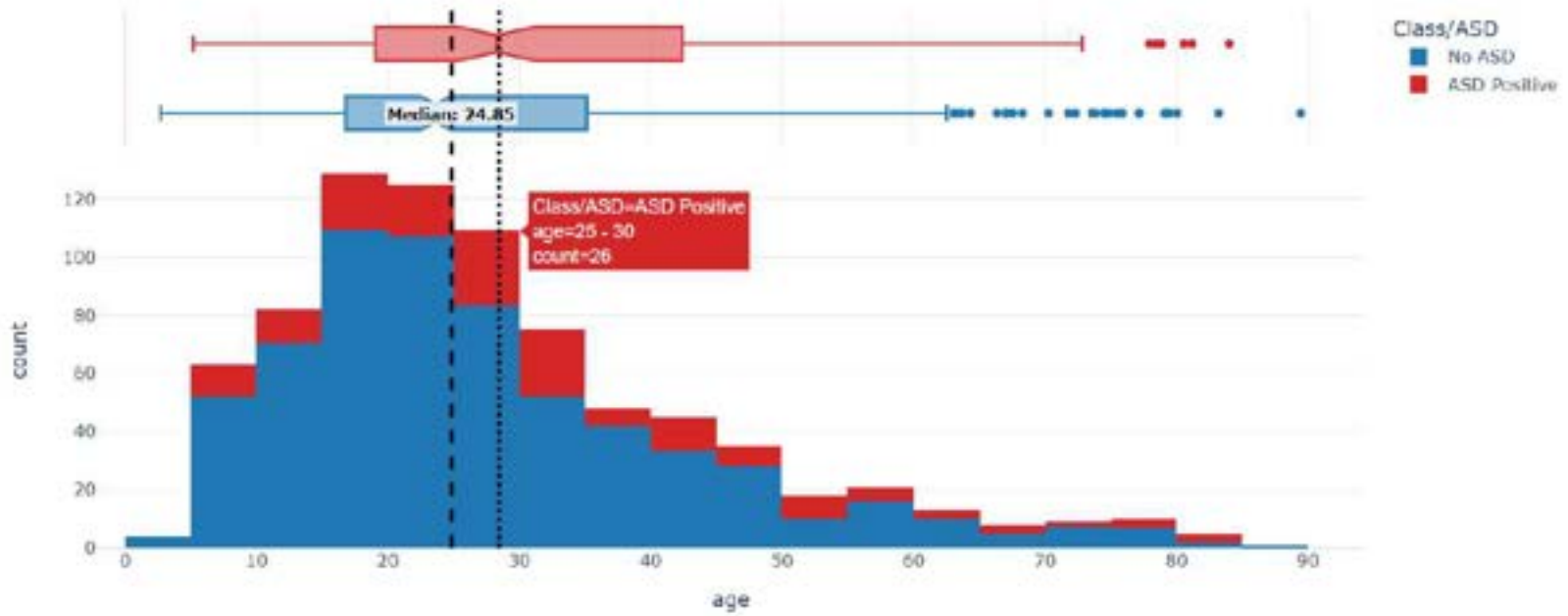
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Age Distribution by ASD Status



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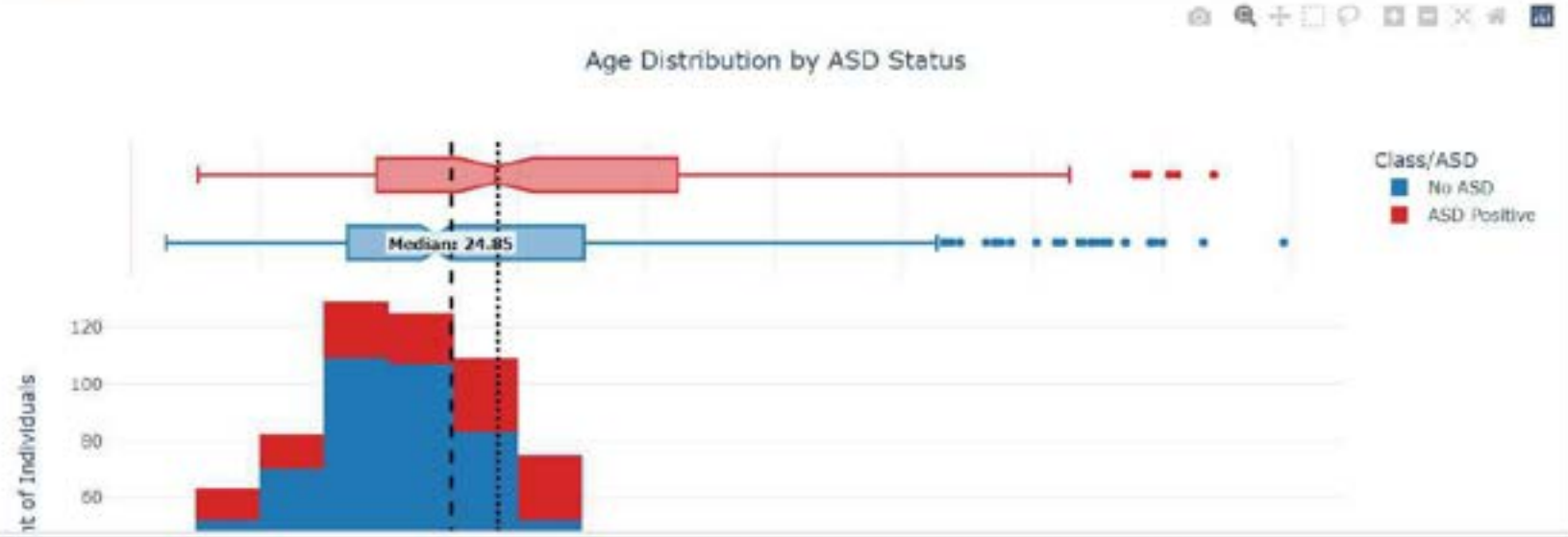
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[30]:
```
# 5. Final layout enhancements
fig.update_layout(
    xaxis_title="Age (Years)",
    yaxis_title="Count of Individuals",
    title_x=0.5,
    hovermode="x unified", # Shows combined tooltips on hover
    showlegend=True
)
fig.show()
```



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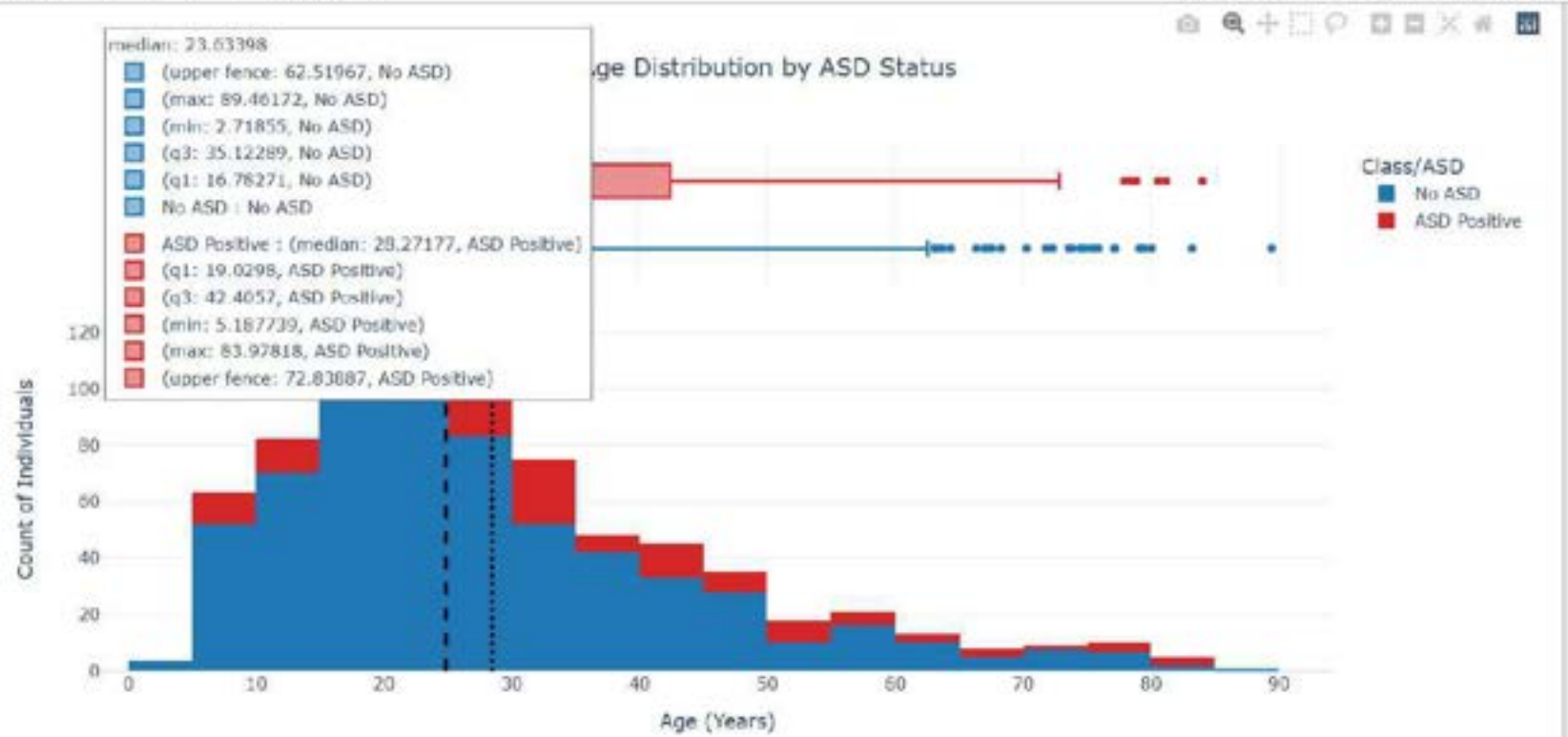
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Code

JupyterLab | Python 3 (pykernel)

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Load the data
df = pd.read_csv("/content/train.csv")

# Set aesthetic style
sns.set(style="whitegrid")

# Create the histogram with color and KDE
plt.figure(figsize=(8,5))
sns.histplot(df["result"], kde=True, color="#1f77b4", bins=20, alpha=0.7)

# Calculate mean and median
result_mean = df["result"].mean()
result_median = df["result"].median()

# Print values
print(f"Mean of result: {result_mean:.2f}")
print(f"Median of result: {result_median:.2f}")

# Add vertical lines for mean and median
plt.axvline(result_mean, color="#e74c3c", linestyle='--', linewidth=2, label=f"Mean = {result_mean:.2f}")
plt.axvline(result_median, color="#27ae60", linestyle='-', linewidth=2, label=f"Median = {result_median:.2f}")

# Titles and labels
plt.title("📊 Distribution of Result", fontsize=14, fontweight='bold', color="#2c3e50")
plt.xlabel("Result Value", fontsize=12)
plt.ylabel("Frequency", fontsize=12)
plt.legend()
plt.tight_layout()
plt.show()
```

```
Mean of result: 8.54
```

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Search

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18:56 06-04-2026

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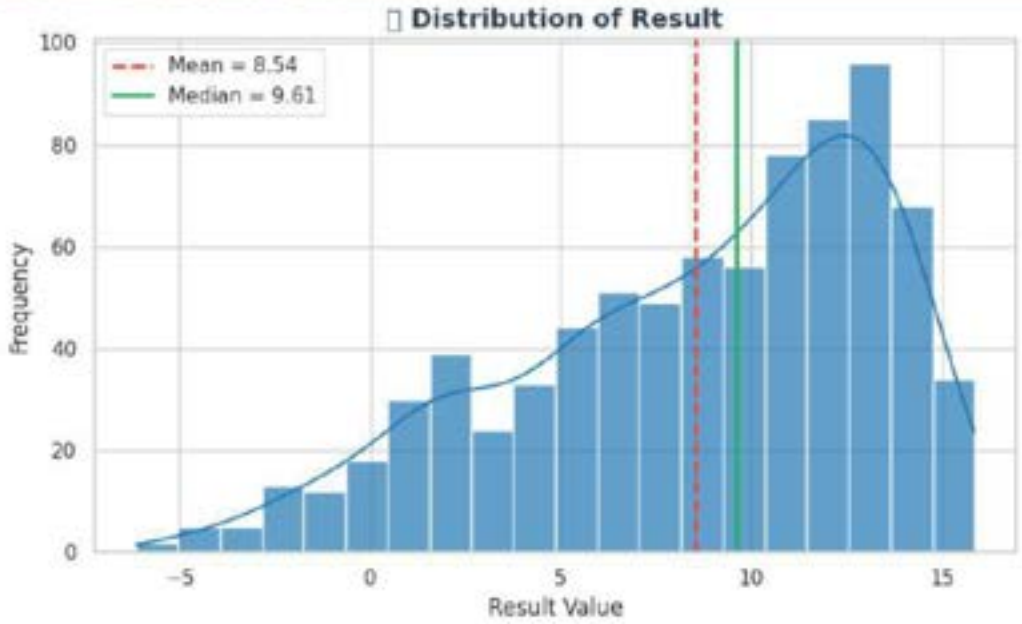
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Not Trusted

Code

JupyterLab | Python 3 (ipykernel)

```
Mean of result: 8.54
Median of result: 9.61
/tmp/ipython-input-2781685255.py:32: UserWarning: Glyph 128202 (\N{BAR CHART}) missing from font(s) DejaVu Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128202 (\N{BAR CHART}) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```



Box plots for identifying outliers in the numerical columns

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Code JupyterLab Python 3 (pykernel)



Box plots for identifying outliers in the numerical columns

[ ]:

```
sns.set(style="whitegrid")

# Create the box plot
plt.figure(figsize=(8,5))
sns.boxplot(
    x=df["age"],
    color="#5b84b0",  # soft teal color
    fliersize=5,      # size of outlier points
    linewidth=2,
    boxprops=dict(facecolor="#a7c7e7", color="#34495e"),  # box fill + edge color
    medianprops=dict(color="#e74c3c", linewidth=2),       # median line color
    whiskerprops=dict(color="#2c3e50", linewidth=1.5)     # whisker color
)

# Add title and labels
plt.title("Box Plot for Age", fontsize=14, fontweight='bold', color="#2c3e50")
plt.xlabel("Age", fontsize=12, color="#2c3e50")

# Show grid and layout
plt.grid(True, linestyle="--", alpha=0.6)
plt.tight_layout()
plt.show()
```

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Home | Autism_Preidiction_using_machin

localhost:8888/notebooks/Desktop/Autism_Preidiction_using_machine_Learningfinal(2).ipynb

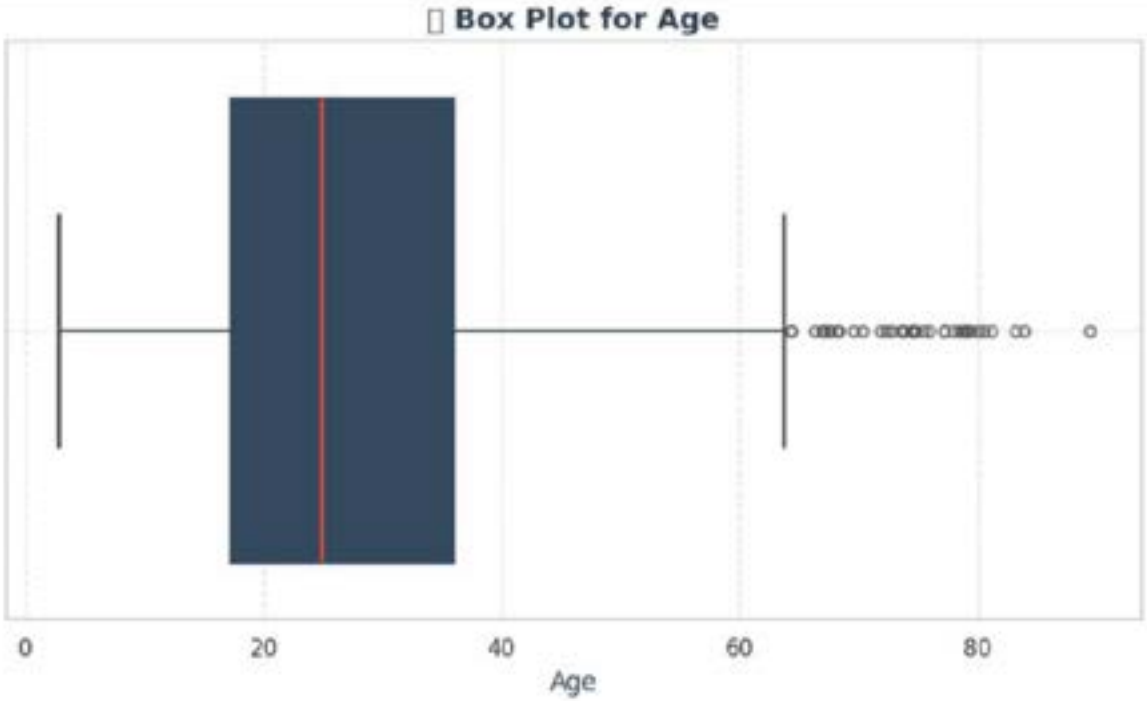
jupyter Autism_Preidiction_using_machine_Learningfinal(2) Last Checkpoint: 41 minutes ago

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Code

JupyterLab | Python 3 (pykernel)




```
sns.set(style="whitegrid")

# Create the box plot
```

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Code

JupyterLab | Python 3 (pykernel)



[ ]:

```
sns.set(style="whitegrid")

# Create the box plot
plt.figure(figsize=(8,5))
sns.boxplot(
    x=df["result"],
    color="#76b5c5",   # soothing teal
    fliersize=5,       # size of outlier points
    linewidth=2,
    boxprops=dict(facecolor="#a8dadc", color="#1d3557"),   # box color & border
    medianprops=dict(color="#e63946", linewidth=2.5),      # median line (red)
    whiskerprops=dict(color="#1d3557", linewidth=1.5),     # whisker color
    capprops=dict(color="#1d3557", linewidth=1.5)          # caps on whiskers
)

# Add title and labels
plt.title("Box Plot for Result", fontsize=14, fontweight='bold', color="#1d3557")
plt.xlabel("Result", fontsize=12, color="#1d3557")

# Grid and layout
plt.grid(True, linestyle="--", alpha=0.6)
plt.tight_layout()
plt.show()
```

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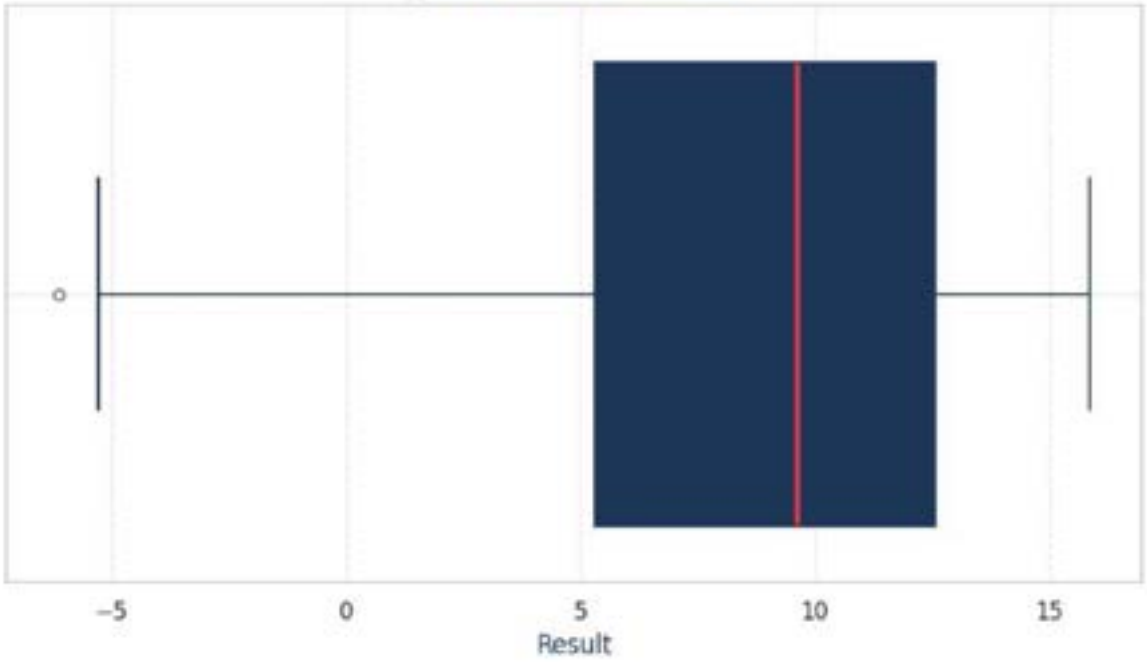
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Code

JupyterLab | Python 3 (pykernel)

Box Plot for Result




```
# count the outliers using IQR method
Q1 = df["age"].quantile(0.25)
Q3 = df["age"].quantile(0.75)
```

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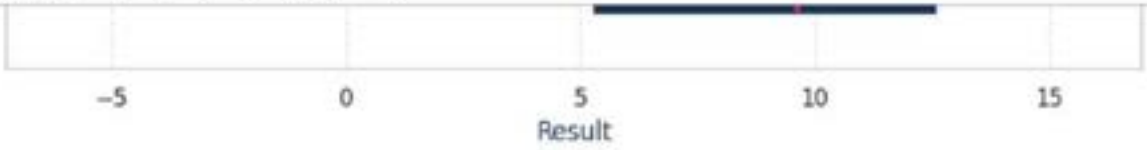
Home | Autism_Preidiction_using_machin | localhost:8888/notebooks/Desktop/Autism_Preidiction_using_machine_Learningfinal(2).ipynb

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Code JupyterLab Python 3 (pykernel)




```
[ ]: # count the outliers using IQR method
Q1 = df["age"].quantile(0.25)
Q3 = df["age"].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
age_outliers = df[(df["age"] < lower_bound) | (df["age"] > upper_bound)]
```

```
[ ]: len(age_outliers)
```

```
[31]: 39
```

```
[ ]: # count the outliers using IQR method
Q1 = df["result"].quantile(0.25)
Q3 = df["result"].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
result_outliers = df[(df["result"] < lower_bound) | (df["result"] > upper_bound)]
```

```
[ ]: len(result_outliers)
```

```
[33]: 1
```

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Code

JupyterLab | Python 3 (pykernel)

Univariate analysis of Categorical columns

```
[ ]: df.columns
```

```
[34]: Index(['A1_Score', 'A2_Score', 'A3_Score', 'A4_Score', 'A5_Score', 'A6_Score',
       'A7_Score', 'A8_Score', 'A9_Score', 'A10_Score', 'age', 'gender',
       'ethnicity', 'jaundice', 'austim', 'contry_of_res', 'used_app_before',
       'result', 'relation', 'Class/ASD'],
      dtype='object')
```

```
[ ]: import matplotlib.pyplot as plt
import seaborn as sns
import random

categorical_columns = [
    'A1_Score', 'A2_Score', 'A3_Score', 'A4_Score', 'A5_Score', 'A6_Score',
    'A7_Score', 'A8_Score', 'A9_Score', 'A10_Score', 'gender', 'ethnicity',
    'jaundice', 'austim', 'contry_of_res', 'used_app_before', 'relation'
]

palettes = ['Set2', 'coolwarm', 'husl', 'cubehelix', 'viridis', 'Spectral', 'pastel', 'mako']

for col in categorical_columns:
    plt.figure(figsize=(8, 5))

    palette_choice = random.choice(palettes)

    sns.countplot(data=df, x=col, palette=palette_choice, edgecolor='black')

    plt.title(f"Count Plot for {col}", fontsize=14, fontweight='bold', color='#333')
```

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Code

JupyterLab | Python 3 (pykernel)

```
import matplotlib.pyplot as plt
import seaborn as sns
import random

categorical_columns = [
    'A1_Score', 'A2_Score', 'A3_Score', 'A4_Score', 'A5_Score', 'A6_Score',
    'A7_Score', 'A8_Score', 'A9_Score', 'A10_Score', 'gender', 'ethnicity',
    'jaundice', 'austim', 'contry_of_res', 'used_app_before', 'relation'
]

palettes = ['Set2', 'coolwarm', 'husl', 'cubehelix', 'viridis', 'Spectral', 'pastel', 'mako']

for col in categorical_columns:
    plt.figure(figsize=(8, 5))

    palette_choice = random.choice(palettes)

    sns.countplot(data=df, x=col, palette=palette_choice, edgecolor='black')

    plt.title(f"Count Plot for {col}", fontsize=14, fontweight='bold', color='#333')
    plt.xlabel(col, fontsize=12)
    plt.ylabel("Count", fontsize=12)


    plt.xticks(rotation=30, ha='right', fontsize=10)
    plt.tight_layout()

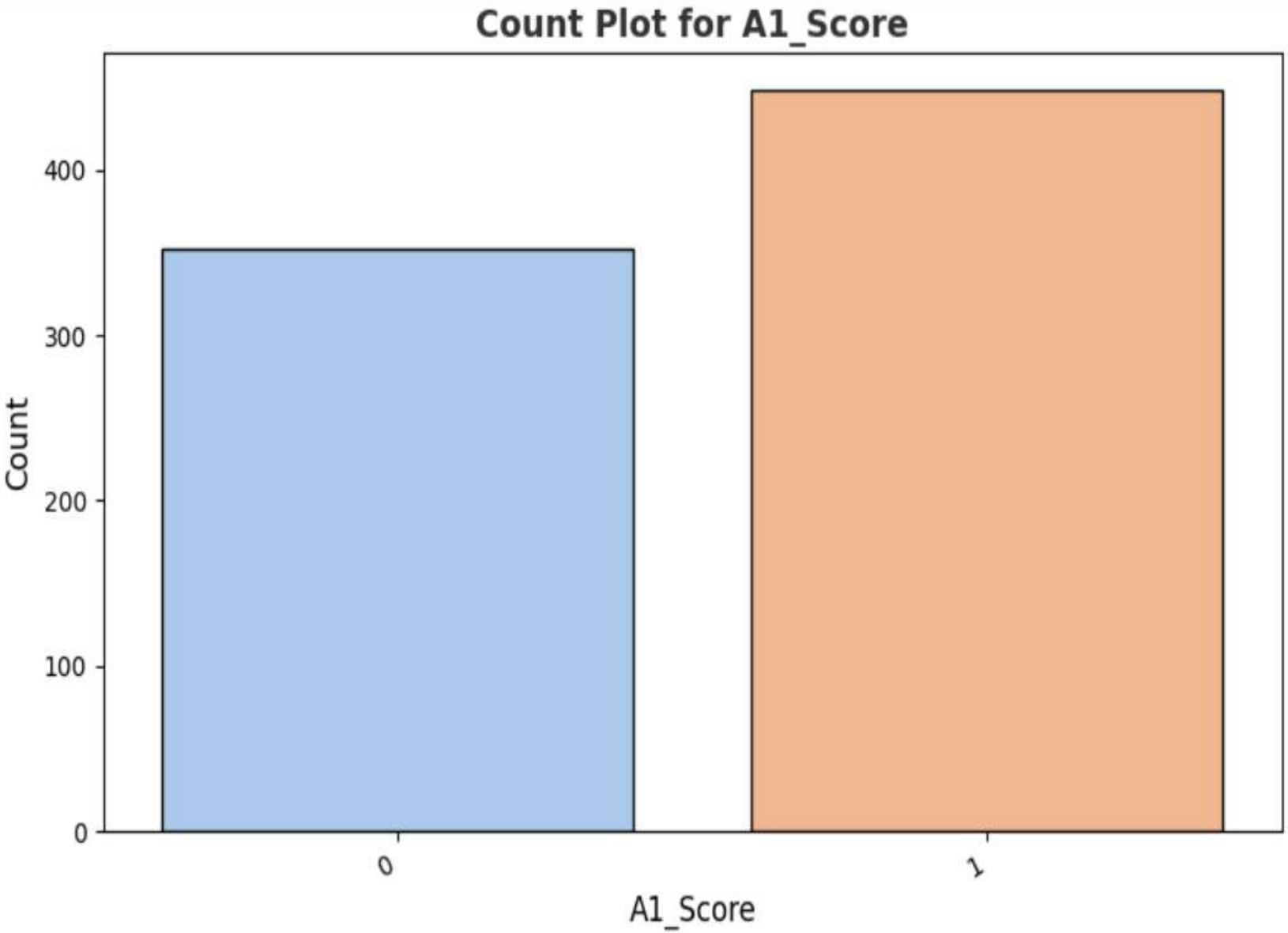
    plt.show()
```

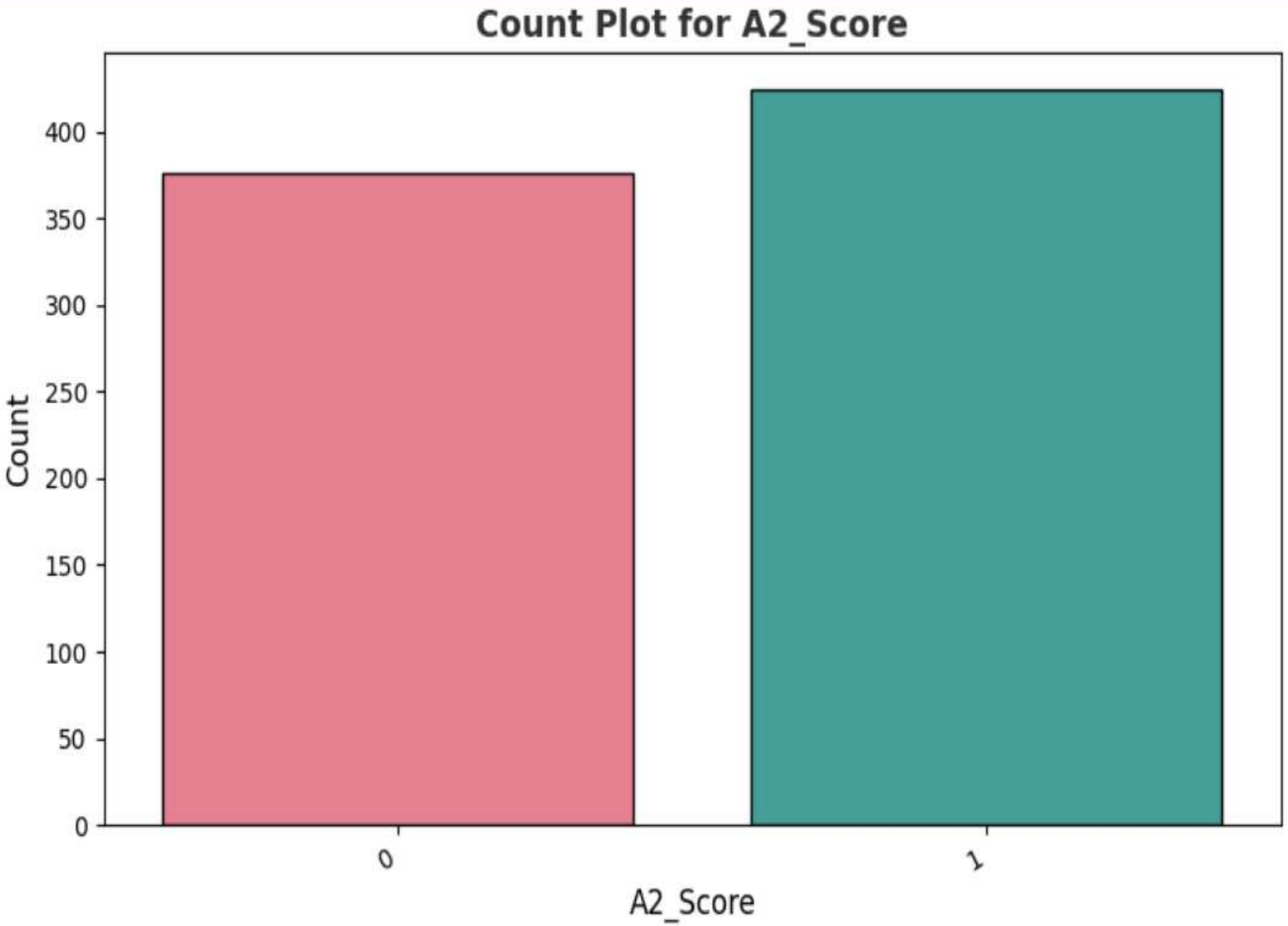
30°C Partly cloudy

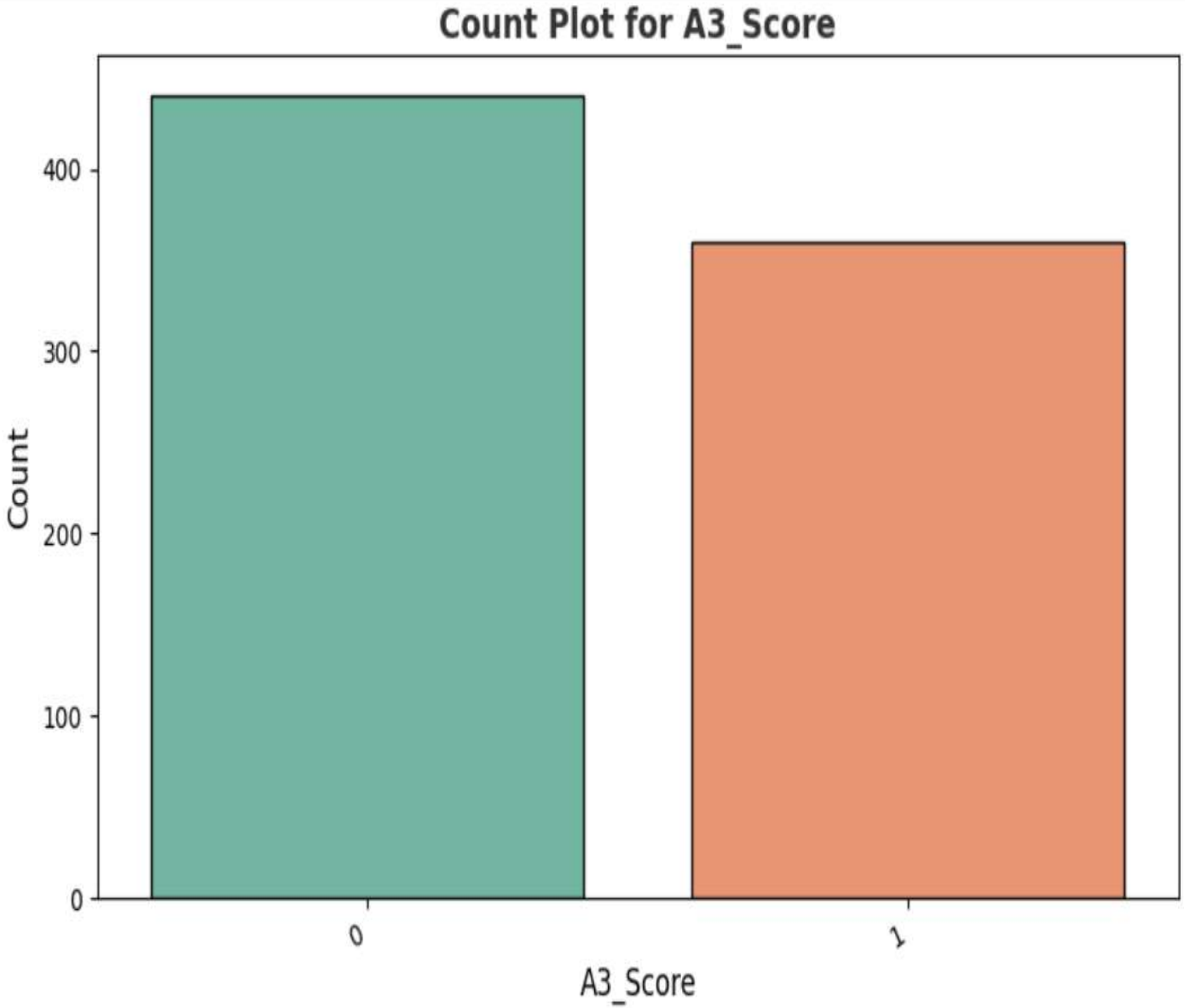
Search

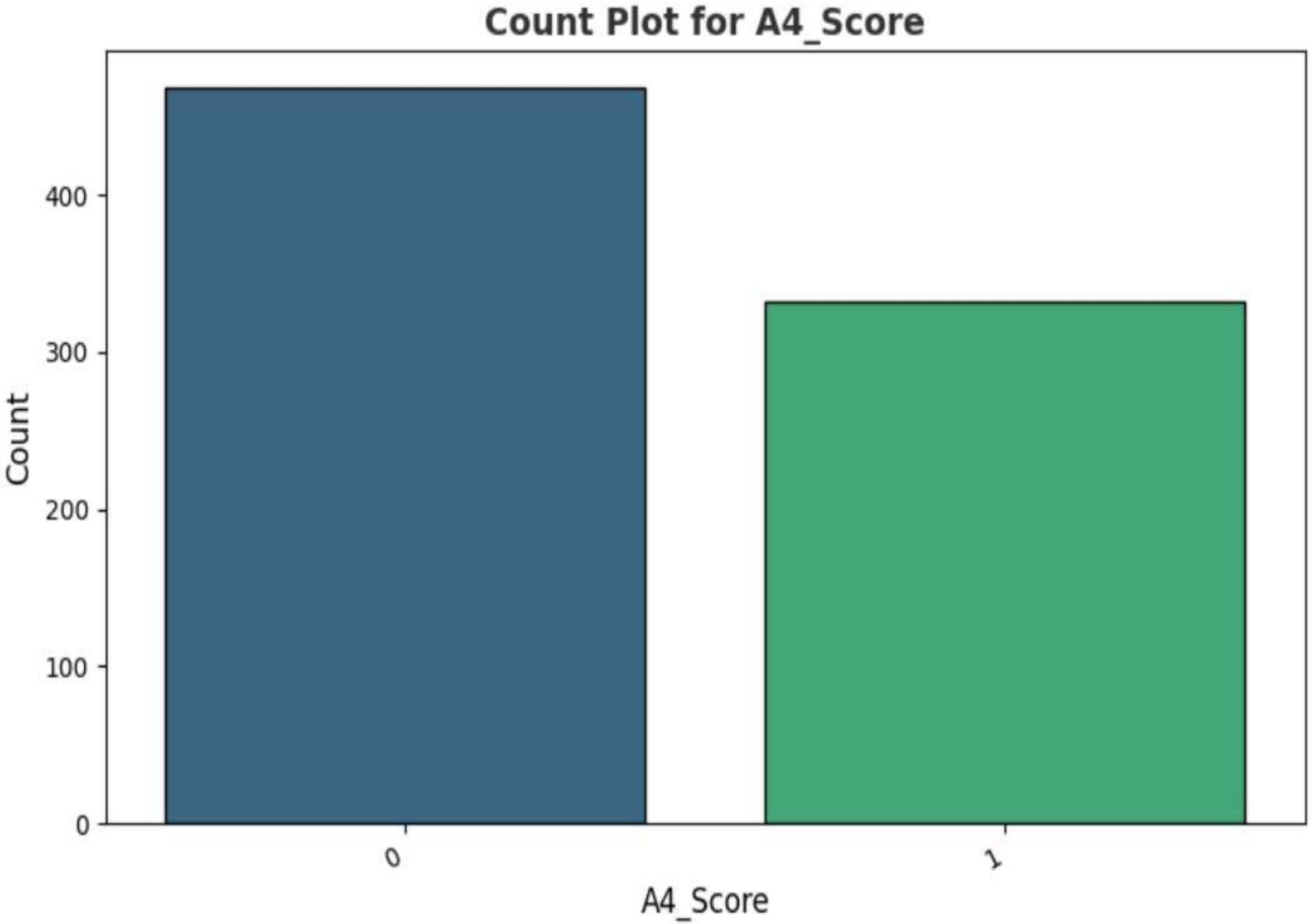
ENG IN

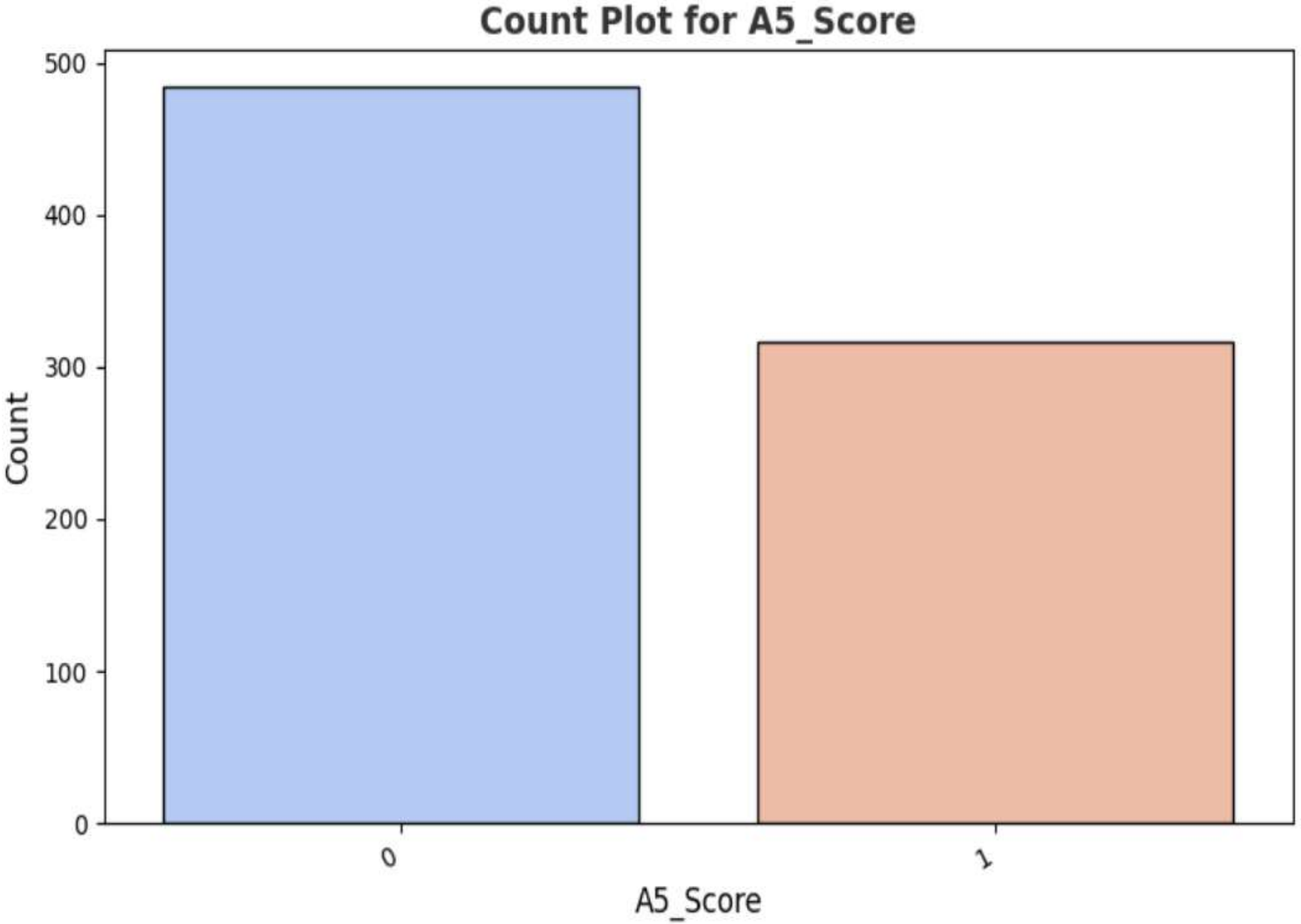
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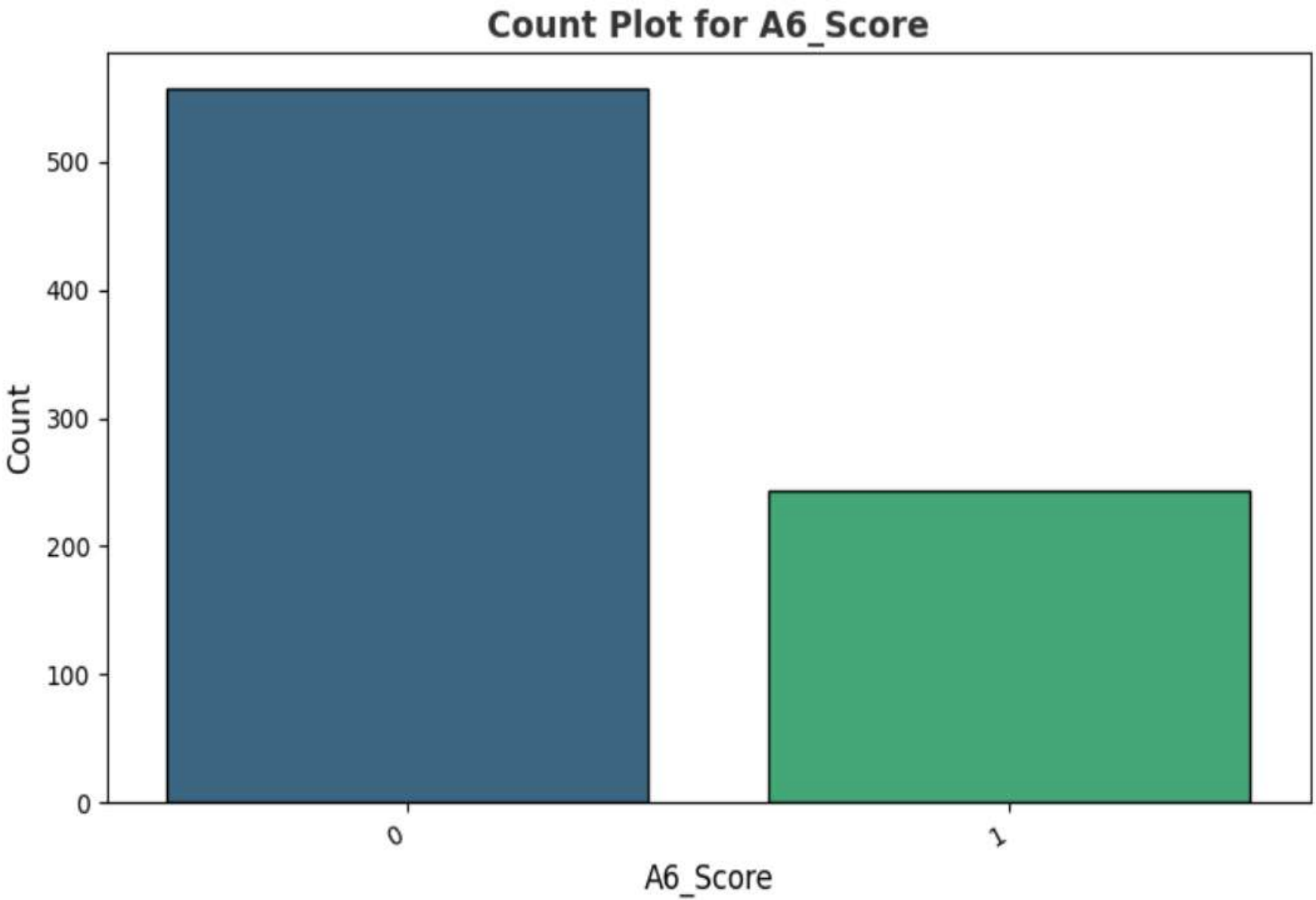

Count Plot for A1_Score
Count
400
300
200
100
0
0
1
A1_Score

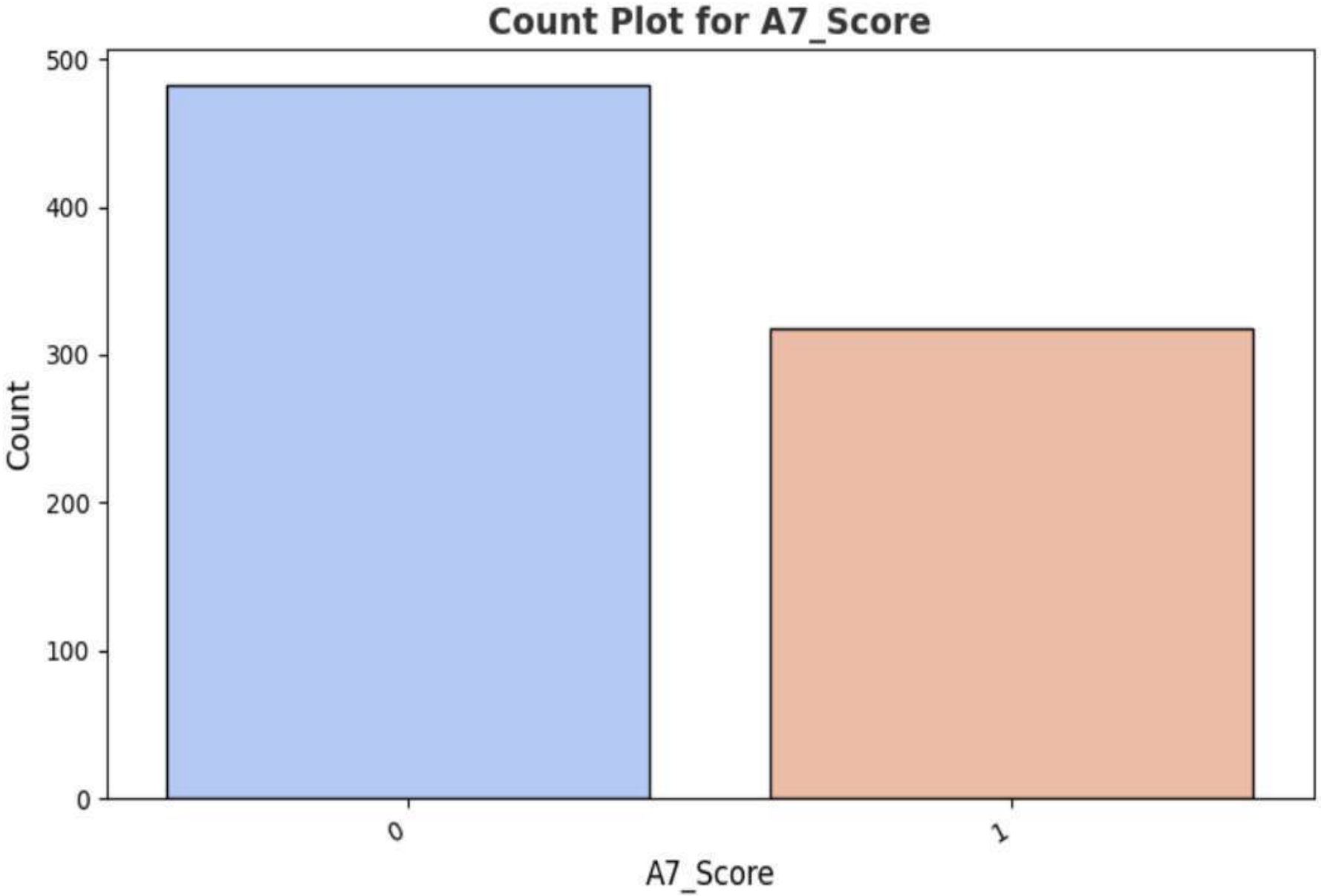

Count Plot for A2_Score
Count
400
350
300
250
200
150
100
50
0
0
1
A2_Score

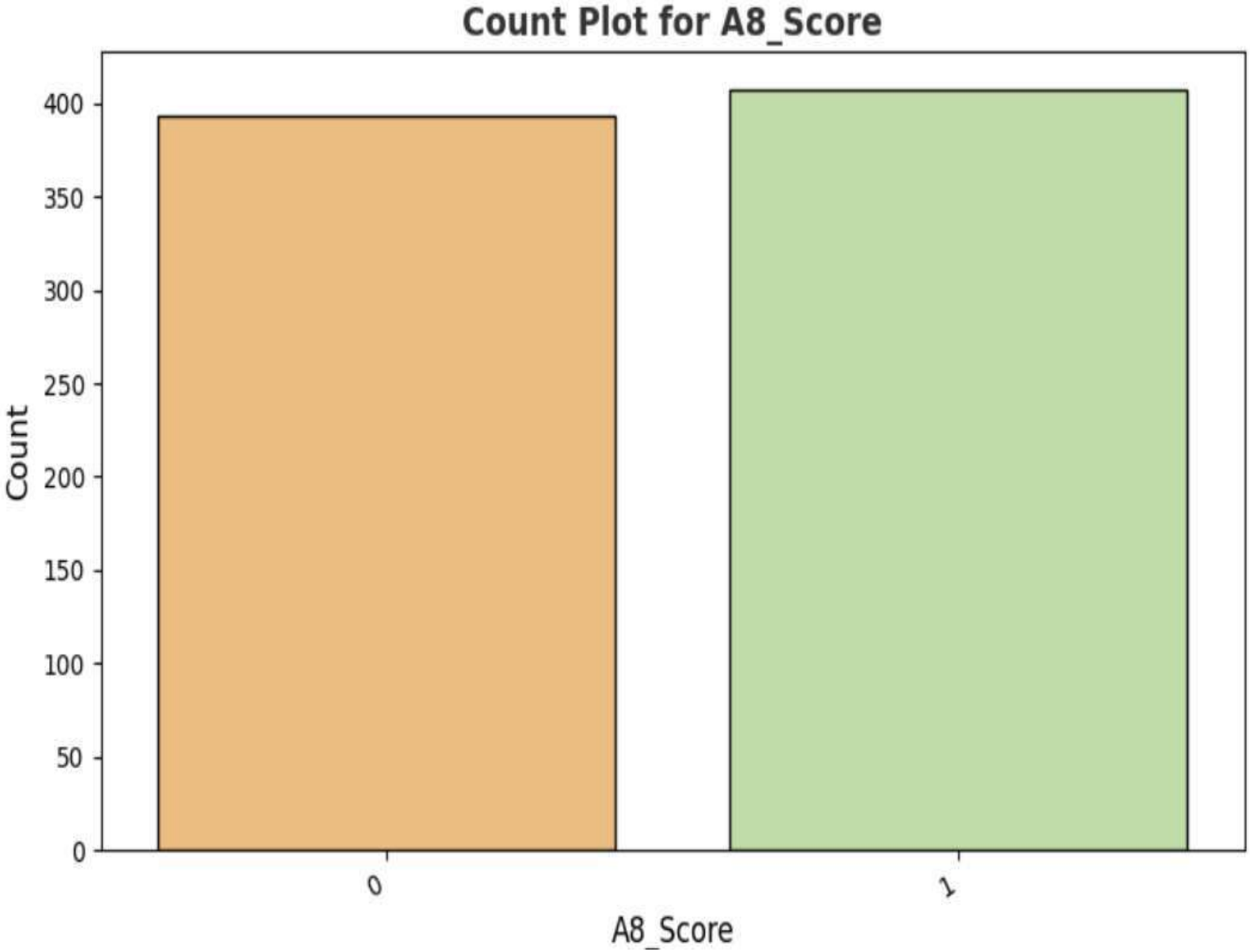

Count Plot for A3_Score
Count
400
300
200
100
0
0
1
A3_Score

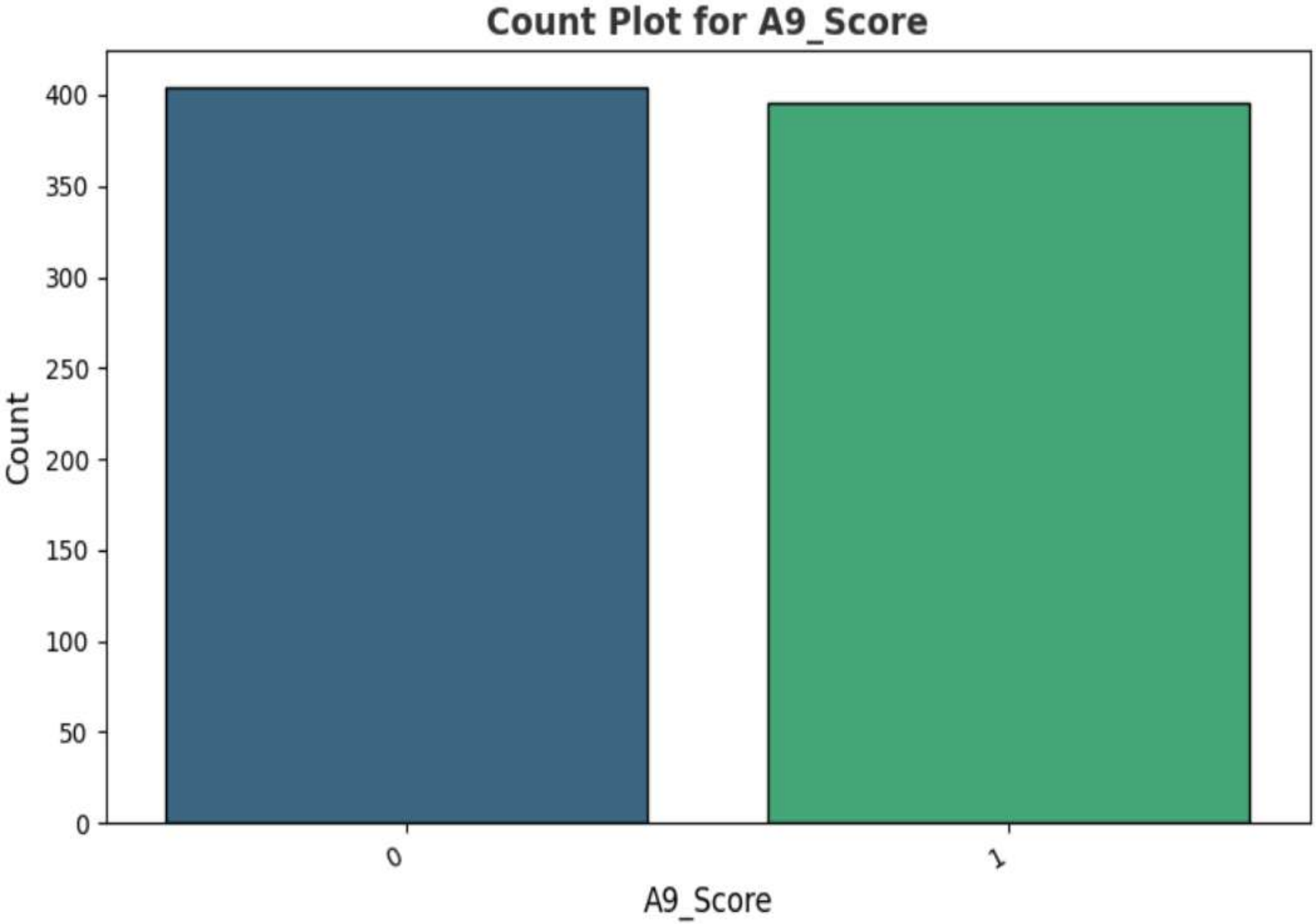

Count Plot for A4_Score
Count
400
300
200
100
0
0
1
A4_Score

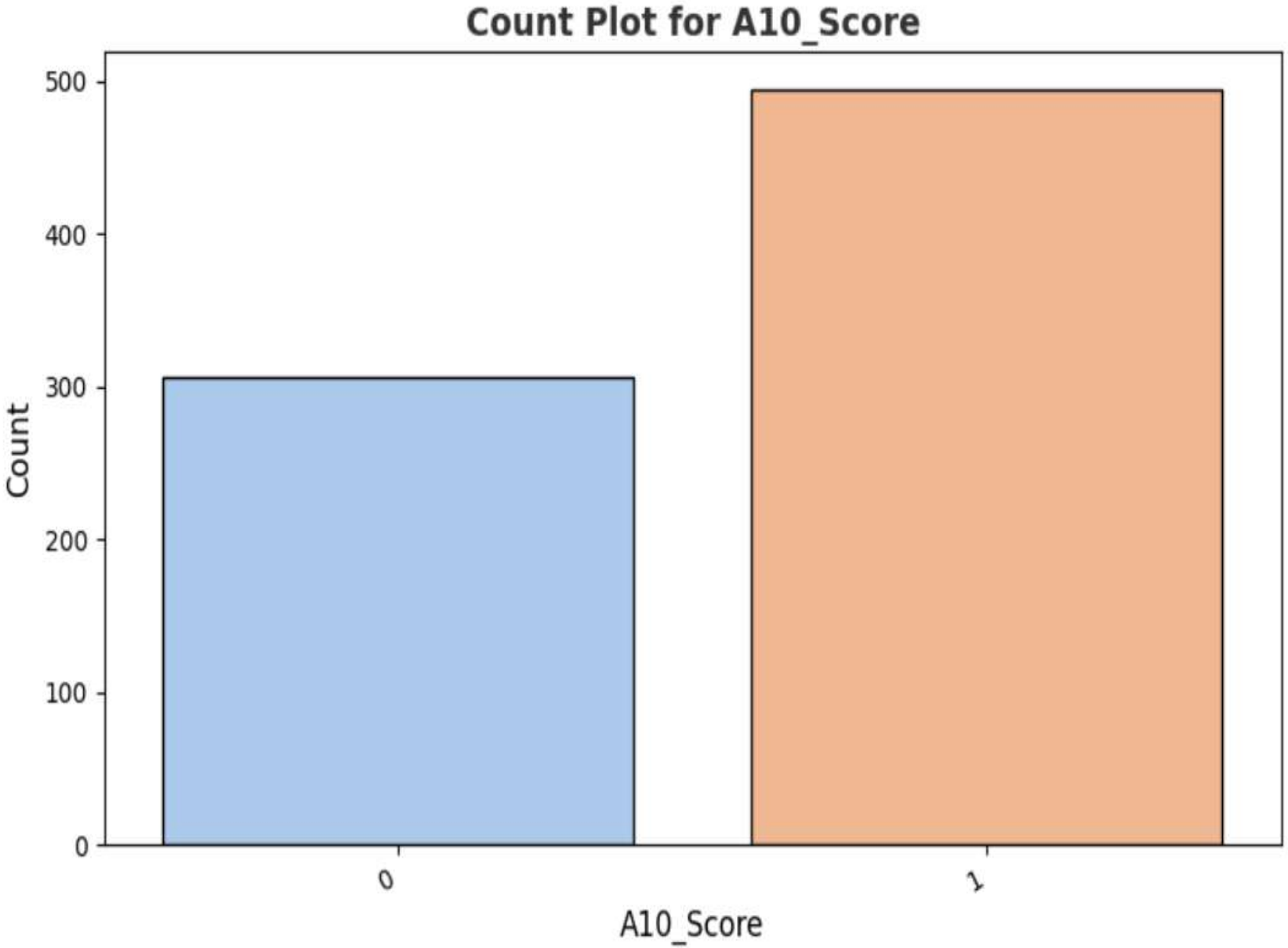

Count Plot for A5_Score
500
400
300
200
100
0
Count
0
1
A5_Score

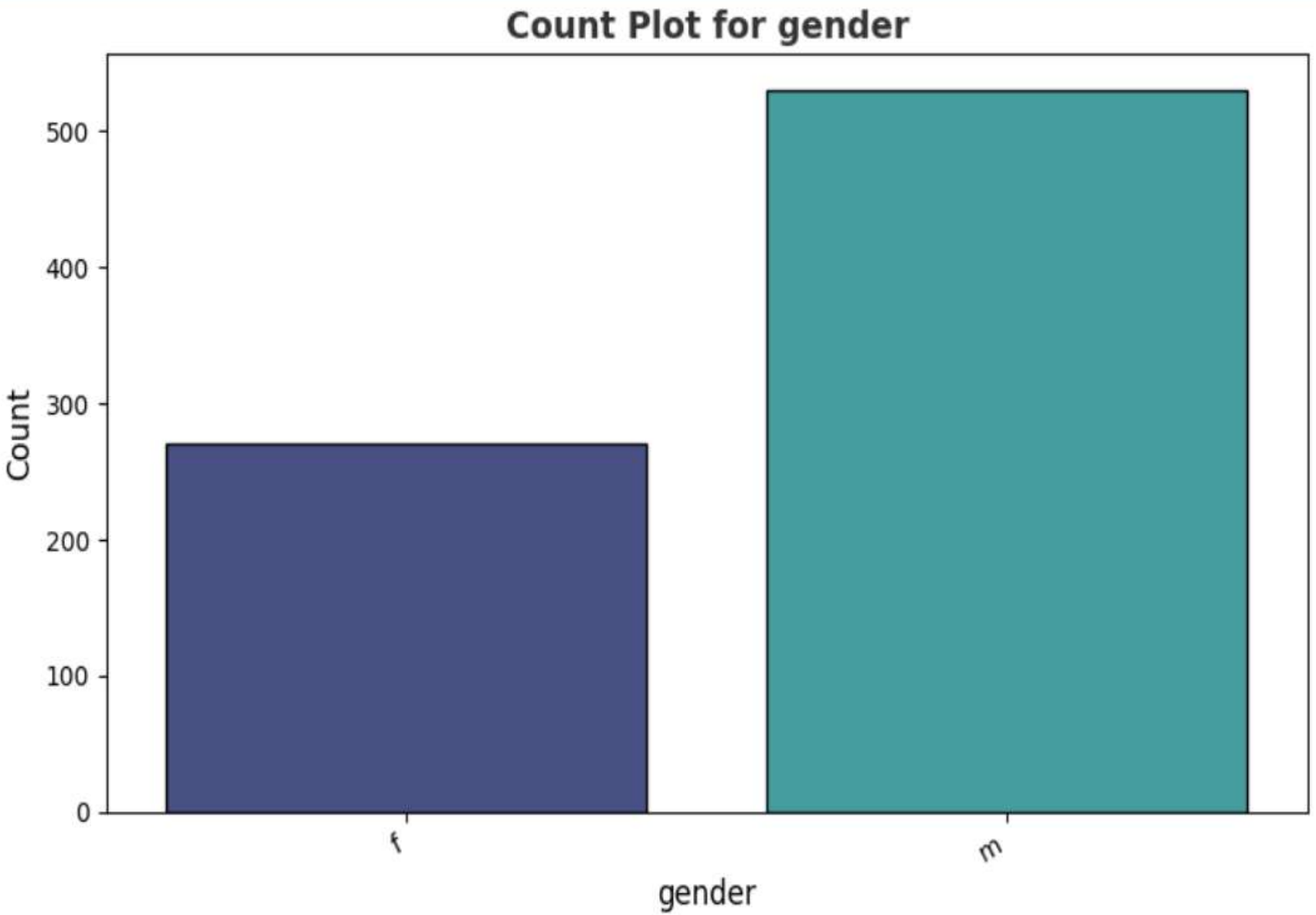

Count Plot for A6_Score
Count
500
400
300
200
100
0
0
1
A6_Score

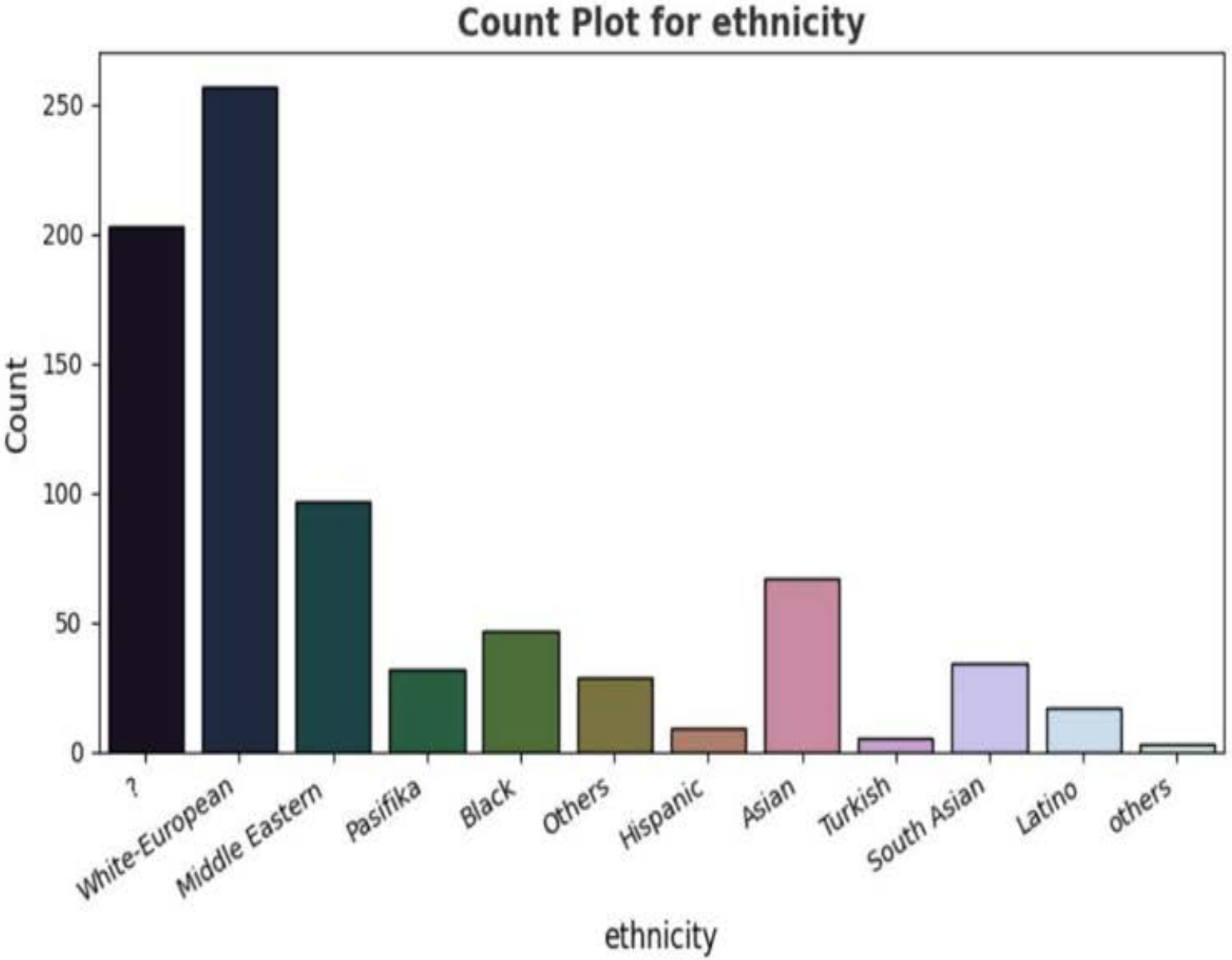

Count Plot for A7_Score
500
400
300
200
100
0
Count
0
1
A7_Score

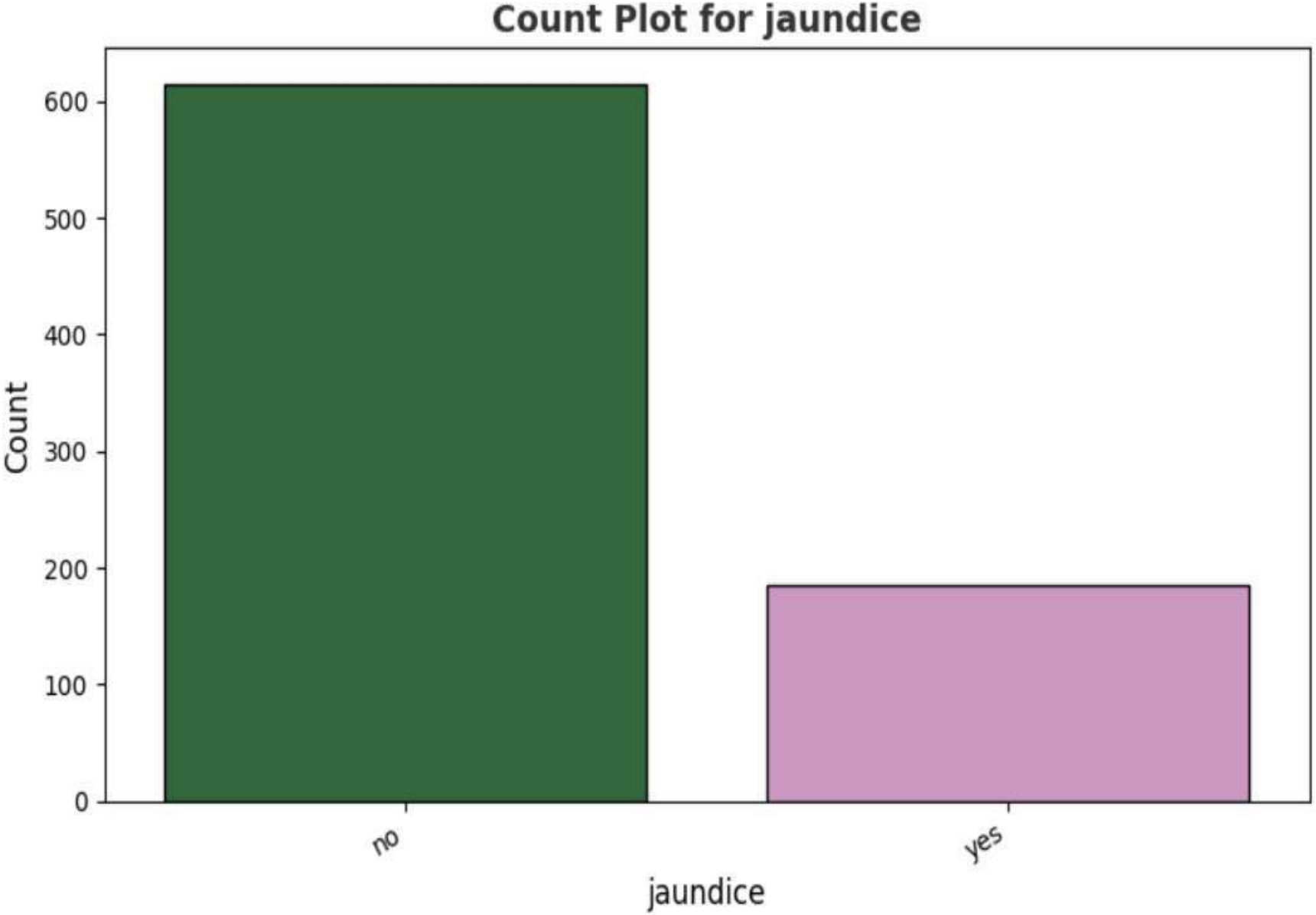

Count Plot for A8_Score
Count
400
350
300
250
200
150
100
50
0
0
1
A8_Score

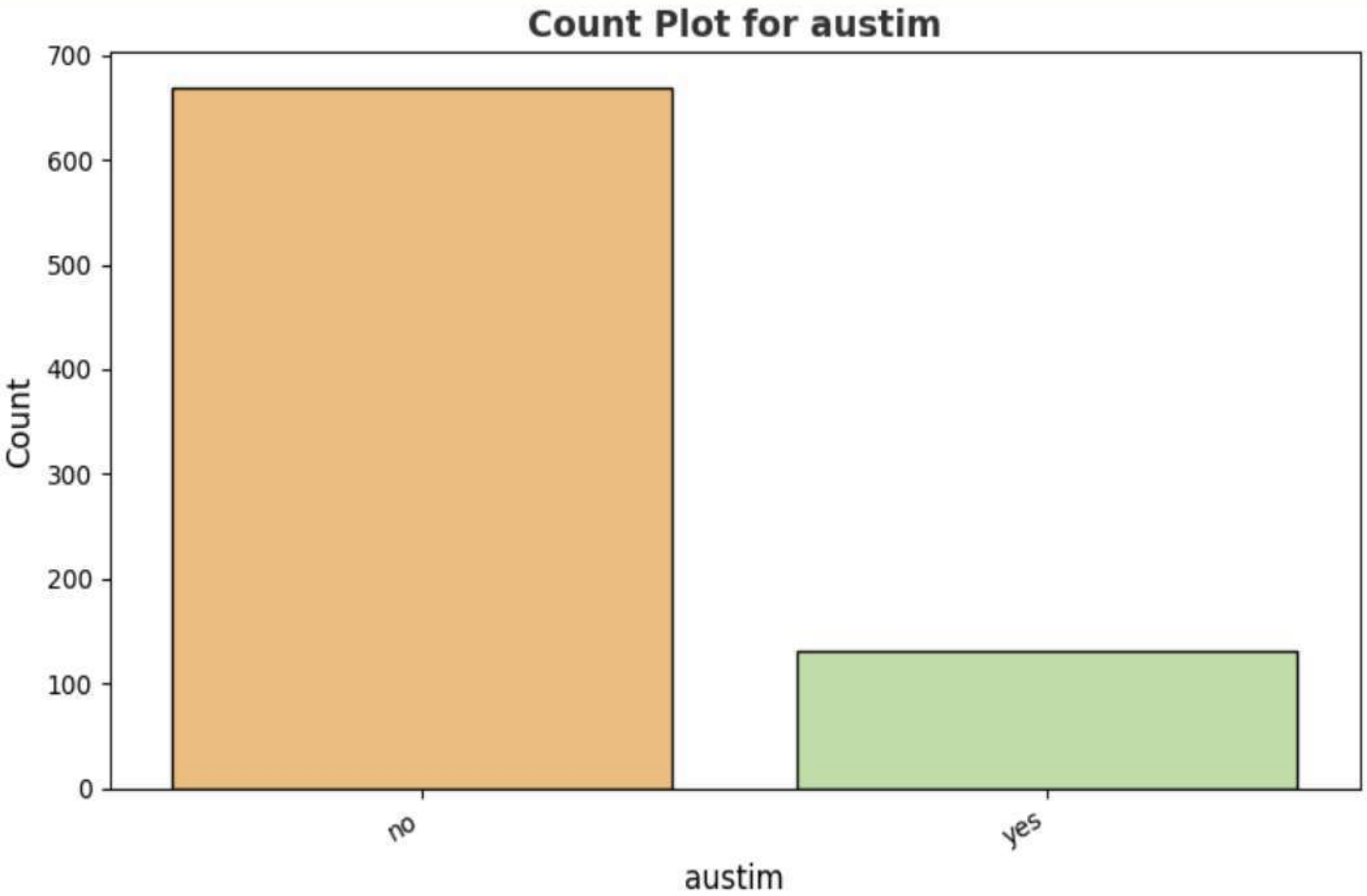

Count Plot for A9_Score
Count
400
350
300
250
200
150
100
50
0
0
1
A9_Score

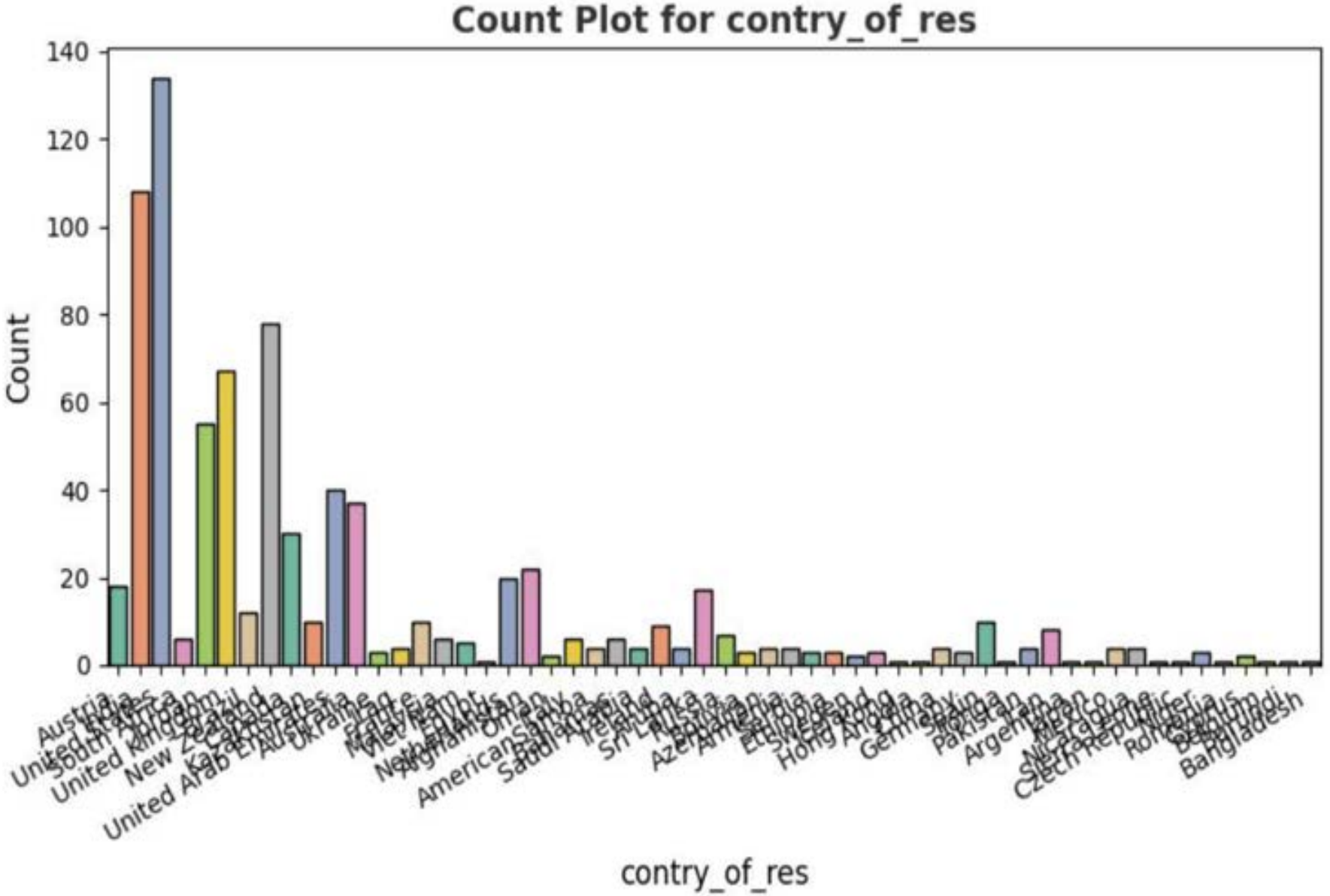

Count Plot for A10_Score
500
400
300
200
100
0
Count
0
1
A10_Score

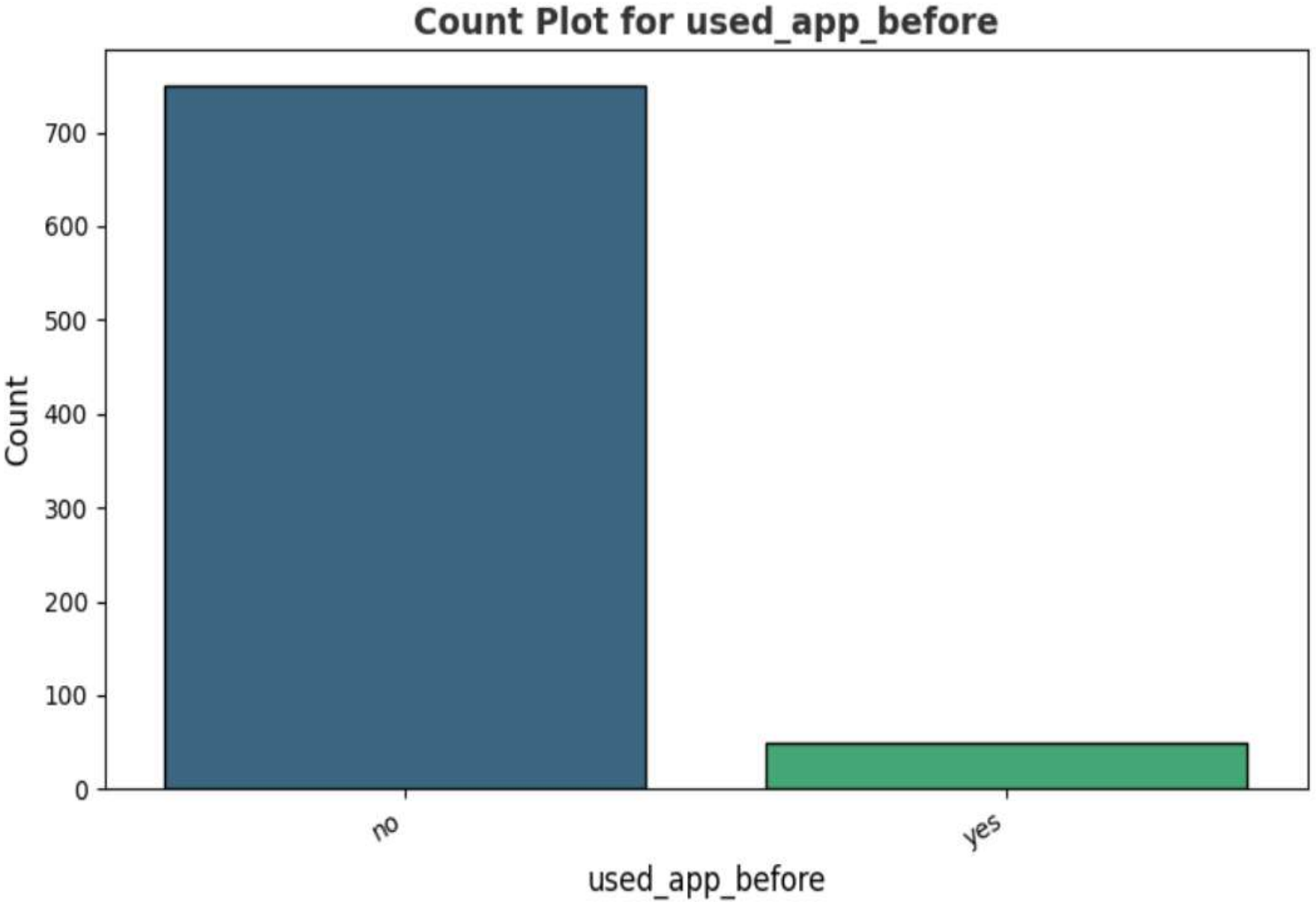

Count Plot for gender
500
400
300
200
100
0
Count
f
m
gender

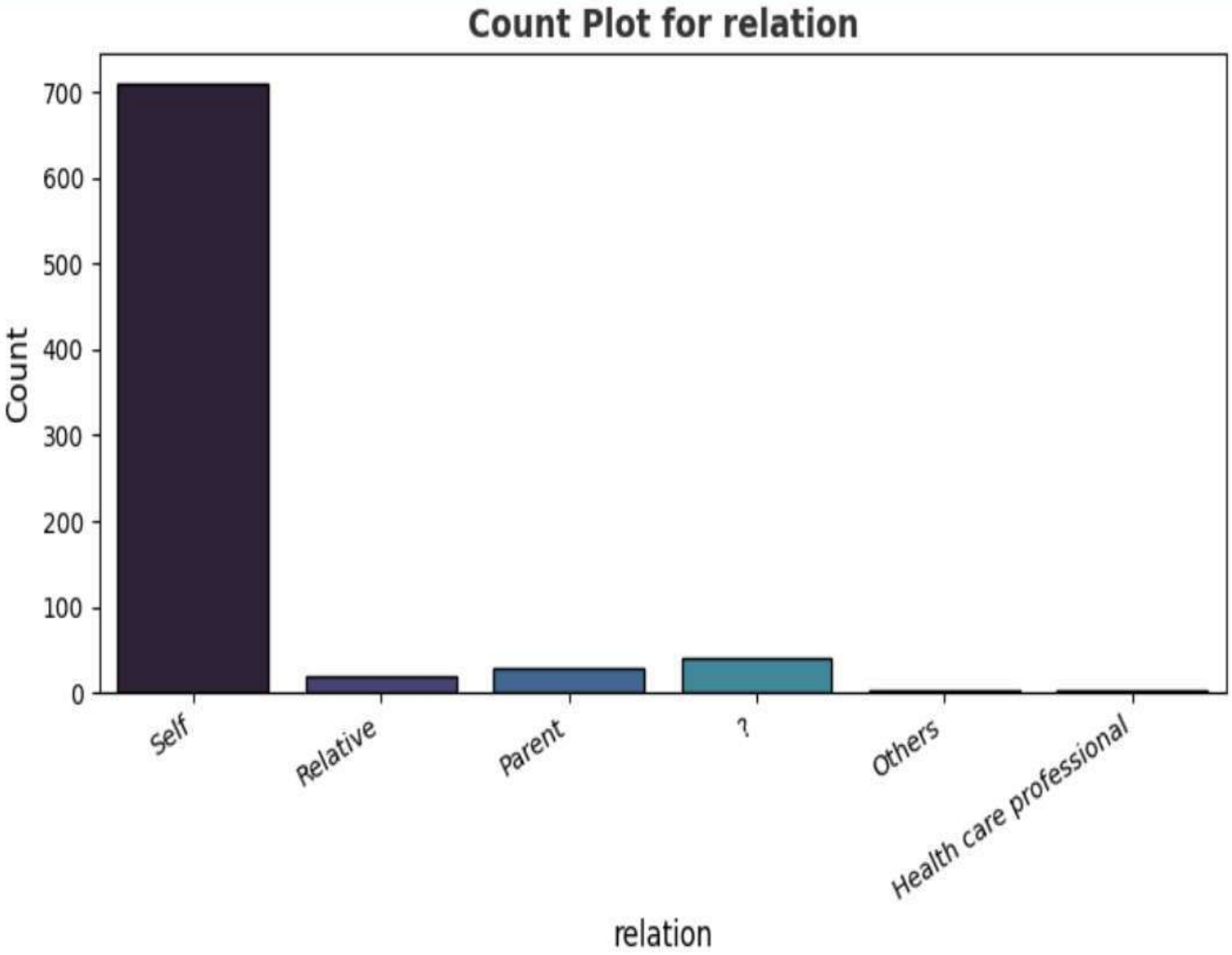

Count Plot for ethnicity
Count
250
200
150
100
50
0
?
White-European
Middle Eastern
Pasifika
Black
Others
Hispanic
Asian
Turkish
South Asian
Latino
others
ethnicity


Count Plot for jaundice
600
500
400
300
200
100
0
Count
no
yes
jaundice


Count Plot for austim
700
600
500
400
300
200
100
0
Count
no
yes
austim


Count Plot for contry_of_res
Count
contry_of_res
0
20
40
60
80
100
120
140


Count Plot for used_app_before
Count
700
600
500
400
300
200
100
0
no
yes
used_app_before


Count Plot for relation
Count
700
600
500
400
300
200
100
0
Self
Relative
Parent
?
Others
Health care professional
relation

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Code

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relation

[ ]:

```
sns.set(style="whitegrid")

# Create figure
plt.figure(figsize=(7,5))

# Create countplot with custom colors
ax = sns.countplot(
    x=df["Class/ASD"],
    palette=["#457b9d", "#e63946"],   # Blue for Non-ASD, red for ASD
    edgecolor="black"
)

# Add labels on bars
for p in ax.patches:
    ax.annotate(
        f'{int(p.get_height())}',
        (p.get_x() + p.get_width() / 2., p.get_height()),
        ha='center', va='bottom',
        fontsize=11, color='black', fontweight='bold'
    )

# Titles and labels
plt.title("Count Plot for Class/ASD", fontsize=14, fontweight='bold', color="#1d3557")
plt.xlabel("Class/ASD (0 = Non-Autistic, 1 = Autistic)", fontsize=12)
plt.ylabel("Count", fontsize=12)
plt.grid(True, linestyle="--", alpha=0.6)
plt.tight_layout()
plt.show()
```

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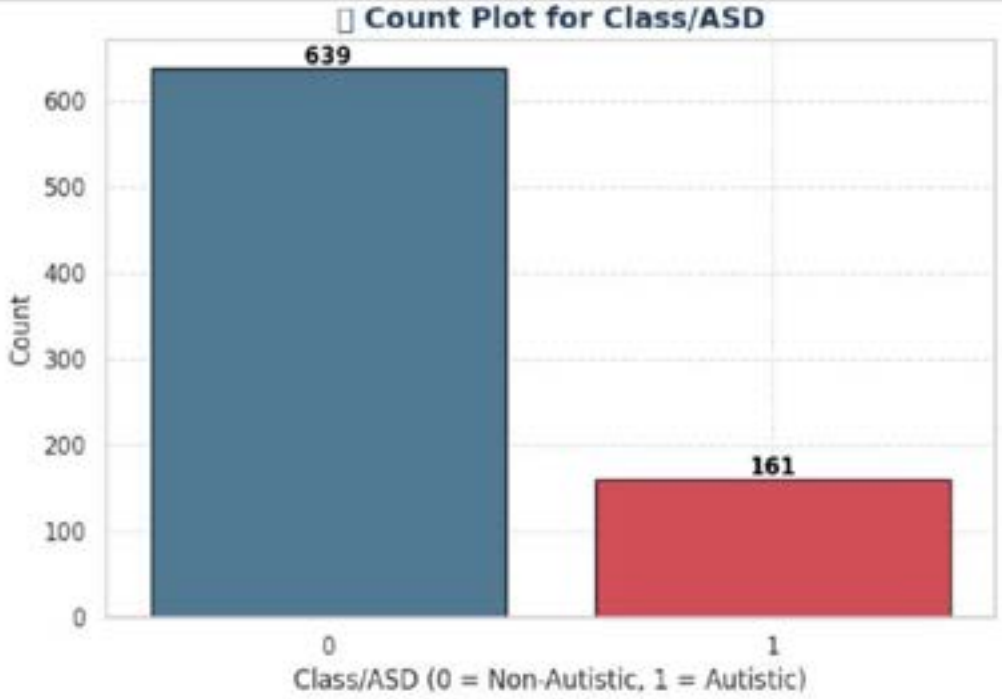
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Code

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```
df["class/ASD"].value_counts()
```

[37]: count

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Code

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```
[ ]: df["Class/ASD"].value_counts()
```

[37]:

Class/ASD	count
0	639
1	161

dtype: int64

handle missing values in ethnicity and relation column

```
[ ]: df["ethnicity"] = df["ethnicity"].replace({"?": "Others", "others": "Others"})
```

```
[ ]: df["ethnicity"].unique()
```

```
[39]: array(['Others', 'White-European', 'Middle Eastern ', 'Pasifika', 'Black',
       'Hispanic', 'Asian', 'Turkish', 'South Asian', 'Latino'],
      dtype=object)
```

```
[ ]: df["relation"].unique()
```

```
[40]: array(['Self', 'Relative', 'Parent', '?', 'Others',
       'Health care professional'], dtype=object)
```

```
[ ]: df["relation"] = df["relation"].replace(
    {"?": "Others",
     "Relative": "Others",
```

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handle missing values in ethnicity and relation column

```
df["ethnicity"] = df["ethnicity"].replace({"?": "Others", "others": "Others"})
```

```
df["ethnicity"].unique()
```

[39]:
```
array(['Others', 'White-European', 'Middle Eastern ', 'Pasifika', 'Black',
       'Hispanic', 'Asian', 'Turkish', 'South Asian', 'Latino'],
      dtype=object)
```

```
df["relation"].unique()
```

[40]:
```
array(['Self', 'Relative', 'Parent', '?', 'Others',
       'Health care professional'], dtype=object)
```

```
df["relation"] = df["relation"].replace(
    {"?": "Others",
     "Relative": "Others",
     "Parent": "Others",
     "Health care professional": "Others"}
)
```

```
df["relation"].unique()
```

[42]:
```
array(['Self', 'Others'], dtype=object)
```

```
df.tail(20)
```

[22]:

	ID	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	cor
780	781	0	1	1	0	0	0	1	1	1	0	23.512162	m	Hispanic	no	no	Us

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```
)
```

[ ]:
```
df["relation"].unique()
```

[42]:
```
array(['Self', 'Others'], dtype=object)
```

[23]:
```
df.tail(20)
```

[24]:

ıre	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	contry_of_res	used_app_before	result	age_desc	relation	Class/ASD
0	1	1	1	0	23.512162	m	Hispanic	no	no	United States	yes	9.220904	18 and more	Relative	ASD Positive
0	1	1	1	1	19.413085	f	?	no	no	Ireland	yes	9.420536	18 and more	Parent	No ASD
0	0	0	0	1	21.541556	m	Asian	no	no	Saudi Arabia	no	13.105263	18 and more	Self	No ASD
1	1	0	1	1	25.531166	f	White-European	no	yes	Bangladesh	no	12.610653	18 and more	Self	No ASD
0	0	0	0	0	19.414728	m	?	no	no	United Arab Emirates	no	9.222105	18 and more	Self	No ASD
1	1	1	1	1	11.275547	m	White-European	no	no	United States	no	7.645019	18 and more	Self	ASD Positive
0	1	1	1	1	37.455881	f	White-European	no	no	United Kingdom	no	12.285082	18 and more	Self	No ASD
1	1	1	1	1	25.148813	m	?	no	yes	United States	no	14.734208	18 and more	Self	No ASD

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Label Encoding

```
[ ]: # Identify columns with "object" data type
object_columns = df.select_dtypes(include=["object"]).columns
```

```
[ ]: print(object_columns)
```

```
Index(['gender', 'ethnicity', 'jaundice', 'austim', 'contry_of_res',
       'used_app_before', 'relation'],
      dtype='object')
```

```
[ ]: # initialize a dictionary to store the encoders
encoders = {}

# apply label encoding and store the encoders
for column in object_columns:
  label_encoder = LabelEncoder()
  df[column] = label_encoder.fit_transform(df[column])
  encoders[column] = label_encoder   # saving the encoder for this column


# save the encoders as a pickle file
with open("encoders.pkl", "wb") as f:
  pickle.dump(encoders, f)
```

```
[ ]: encoders
```

```
[47]: {'gender': LabelEncoder(),
 'ethnicity': LabelEncoder(),
 'jaundice': LabelEncoder(),
 'austim': LabelEncoder()
```

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```
[ ]: encoders
```

```
[47]: {'gender': LabelEncoder(),
 'ethnicity': LabelEncoder(),
 'jaundice': LabelEncoder(),
 'austim': LabelEncoder(),
 'contry_of_res': LabelEncoder(),
 'used_app_before': LabelEncoder(),
 'relation': LabelEncoder()}
```

```
[ ]: df.head()
```

[48]:

A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	contry_of_res	used_app_before	result	relation	Class/ASD
1	0	1	0	1	1	38	0	5	0	0	6	0	6.351166	1	0
0	0	0	0	0	0	47	1	5	0	0	23	0	2.255185	1	0
1	1	1	1	1	1	7	1	9	0	1	52	0	14.851484	1	1
0	0	0	0	0	0	23	0	5	0	0	52	0	2.276617	1	0
0	0	0	0	0	0	43	1	5	0	0	44	0	-4.777286	1	0

```
[ ]:
# Select only numeric columns for correlation
numeric_df = df.select_dtypes(include=["number"])

# Compute correlation matrix
corr_matrix = numeric_df.corr()
```

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Code

JupyterLab | Python 3 (pykernel)

[23]:
```
# Select only numeric columns for correlation
numeric_df = df.select_dtypes(include=["number"])

# Compute correlation matrix
corr_matrix = numeric_df.corr()

# Create interactive heatmap using Plotly
fig = px.imshow(
    corr_matrix,
    text_auto=".2f",  # show values inside cells
    color_continuous_scale="Tealrose",  # blue-red color scale
    title="Correlation Heatmap of Numeric Features",
    aspect="auto"
)

# Update layout for better readability
fig.update_layout(
    width=800,
    height=600,
    title_font=dict(size=18, family="Arial", color="#1d3557"),
    xaxis_title="Features",
    yaxis_title="Features",
    coloraxis_colorbar=dict(title="Correlation"),
)

# Show interactive plot
fig.show()
```

Correlation Heatmap of Numeric Features

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Chat

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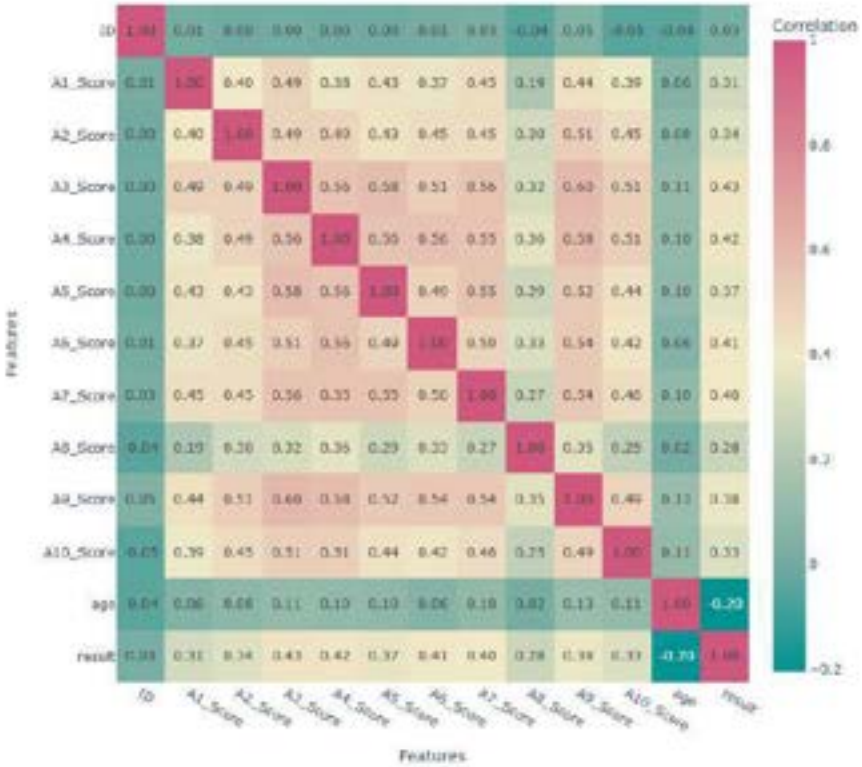
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Code

JupyterLab Python 3 (ipykernel)

Correlation Heatmap of Numeric Features



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Bivariate Analysis

Insights from EDA:

- There are few outliers in the numerical columns (age, results)
- There is a class imbalance in the target column
- There is a class imbalance in the categorical features
- We don't have any highly correlated column
- performed label encoding and saved the encoders

4. Data preprocessing

Handling teh outliers

```
# function to replace the outliers with median
def replace_outliers_with_median(df, column):
  Q1 = df[column].quantile(0.25)
  Q3 = df[column].quantile(0.75)
  IQR = Q3 - Q1

  lower_bound = Q1 - 1.5 * IQR
  upper_bound = Q3 + 1.5 * IQR

  median = df[column].median()

  # replace outliers with median value
  df[column] = df[column].apply(lambda x: median if x < lower_bound or x > upper_bound else x)

  return df
```

```
# replace outliers in the "age" column
df = replace_outliers_with_median(df, "age")

# replace outliers in the "result" column
df = replace_outliers_with_median(df, "result")
```

```
df.head()
```

	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	contry_of_res
0	1	0	1	0	1	0	1	0	1	1	18.0	0	5	0	0	4

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Code | JupyterLab | Python 3 (ipykernel)

4. Data preprocessing

Handling teh outliers

```
# function to replace the outliers with median
def replace_outliers_with_median(df, column):
  Q1 = df[column].quantile(0.25)
  Q3 = df[column].quantile(0.75)
  IQR = Q3 - Q1

  lower_bound = Q1 - 1.5 * IQR
  upper_bound = Q3 + 1.5 * IQR

  median = df[column].median()

  # replace outliers with median value
  df[column] = df[column].apply(lambda x: median if x < lower_bound or x > upper_bound else x)

  return df
```

```
# replace outliers in the "age" column
df = replace_outliers_with_median(df, "age")

# replace outliers in the "result" column
df = replace_outliers_with_median(df, "result")
```

```
df.head()
```

[52]:

	A1_Score	A2_Score	A3_Score	A4_Score	A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	contry_of_res	s
0	1	0	1	0	1	0	1	0	1	1	38.0	0	5	0	0	6	
1	0	0	0	0	0	0	0	0	0	0	47.0	1	5	0	0	23	
2	1	1	1	1	1	1	1	1	1	1	7.0	1	9	0	1	52	

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```
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    median = df[column].median()

    # replace outliers with median value
    df[column] = df[column].apply(lambda x: median if x < lower_bound or x > upper_bound else x)

    return df
```

[ ]:
```
# replace outliers in the "age" column
df = replace_outliers_with_median(df, "age")

# replace outliers in the "result" column
df = replace_outliers_with_median(df, "result")
```

[ ]:
```
df.head()
```

[52]:

A5_Score	A6_Score	A7_Score	A8_Score	A9_Score	A10_Score	age	gender	ethnicity	jaundice	austim	contry_of_res	used_app_before	result	relation	Class/ASD
1	0	1	0	1	1	38.0	0	5	0	0	6	0	6.351166	1	0
0	0	0	0	0	0	47.0	1	5	0	0	23	0	2.255185	1	0
1	1	1	1	1	1	7.0	1	9	0	1	52	0	14.851484	1	1
0	0	0	0	0	0	23.0	0	5	0	0	52	0	2.276617	1	0
0	0	0	0	0	0	43.0	1	5	0	0	44	0	-4.777286	1	0

[ ]:
```
df.shape
```

[53]:
```
(800, 20)
```

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Train Test Split

```
df.columns
```

```
Index(['A1_Score', 'A2_Score', 'A3_Score', 'A4_Score', 'A5_Score', 'A6_Score',
       'A7_Score', 'A8_Score', 'A9_Score', 'A10_Score', 'age', 'gender',
       'ethnicity', 'jaundice', 'austim', 'contry_of_res', 'used_app_before',
       'result', 'relation', 'Class/ASD'],
      dtype='object')
```

```
X = df.drop(columns=["Class/ASD"])
y = df["Class/ASD"]
```

```
print(X)
```

```
     A1_Score  A2_Score  A3_Score  A4_Score  A5_Score  A6_Score  A7_Score  \
0           1         0         1         0         1         0         1
1           0         0         0         0         0         0         0
2           1         1         1         1         1         1         1
3           0         0         0         0         0         0         0
4           0         0         0         0         0         0         0
..        ...       ...       ...       ...       ...       ...       ...
795         0         1         0         0         0         0         0
796         0         1         1         0         0         1         0
797         0         0         0         0         0         0         0
798         0         0         0         0         0         0         0
799         0         1         0         0         0         0         0

     A8_Score  A9_Score  A10_Score   age  gender  ethnicity  jaundice  austim  \
0           0         1          1  58.0       0          5         0       0
1           0         0          0  47.0       1          5         0       0
2           1         1          1   7.0       1          9         0       1
3           0         0          0  23.0       0          5         0       0
4           0         0          0  43.0       1          5         0       0
..        ...       ...        ...   ...     ...        ...       ...     ...
795         0         1          1  16.0       1          2         0       0
```

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Code

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```
     AR_Score  A9_Score  A10_Score   age  gender  ethnicity  jaundice  austim  \
0           0         1          1  38.0       0          5         0       0
1           0         0          0  47.0       1          5         0       0
2           1         1          1   7.0       1          9         0       1
3           0         0          0  23.0       0          5         0       0
4           0         0          0  43.0       1          5         0       0
..        ...       ...        ...   ...     ...        ...       ...     ...
795         0         1          1  16.0       1          2         0       0
796         1         1          1  20.0       1          9         0       0
797         0         0          0   5.0       1          7         1       0
798         0         0          0  16.0       0          5         0       0
799         0         0          0  46.0       0          5         0       0

     contry_of_res  used_app_before     result  relation
0                6                0   0.351106         1
1               25                0   2.255185         1
2               52                0  14.851484         1
3               52                0   2.276617         1
4               44                0  -4.777286         1
..             ...              ...        ...       ...
795             34                0  12.999501         1
796             16                0  13.561518         1
797             34                0   2.653177         1
798             14                0   0.069942         1
799             50                1   2.243304         1

[800 rows x 19 columns]
```

[ ]:
```
print(y)
```

```
0      0
1      0
2      1
3      0
4      0
      ..
795    0
796    0
```

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[ ]:
```
print(y)
```

```
0      0
1      0
2      1
3      0
4      0
      ..
795    0
796    0
797    0
798    0
799    0
Name: Class/ASD, Length: 800, dtype: int64
```

[ ]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

[ ]:
```
print(y_train.shape)
print(y_test.shape)
```

```
(640,)
(160,)
```

[ ]:
```
y_train.value_counts()
```

[60]:

Class/ASD	count
0	515
1	125

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Code | JupyterLab | Python 3 (pykernel)

```
[ ]: y_train.value_counts()
```

[60]:

Class/ASD	count
0	515
1	125

dtype: int64

```
[ ]: y_test.value_counts()
```

[61]:

Class/ASD	count
0	124
1	36

dtype: int64

SMOTE (Synthetic Minority Oversampling technique)

```
[ ]: smote = SMOTE(random_state=42)
```

```
[ ]: X_train_smote, y_train_smote = smote.fit_resample(X_train, y_train)
```

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Code JupyterLab Python 3 (pykernel)

SMOTE (Synthetic Minority Oversampling technique)

```
smote = SMOTE(random_state=42)
```

```
X_train_smote, y_train_smote = smote.fit_resample(X_train, y_train)
```

```
print(y_train_smote.shape)
```

```
(1030,)
```

```
print(y_train_smote.value_counts())
```

```
Class/ASD
1    515
0    515
Name: count, dtype: int64
```

```
from imblearn.over_sampling import SMOTE

sm = SMOTE(random_state=42)
X_res, y_res = sm.fit_resample(X_scaled, y)
print("After balancing:", np.bincount(y_res))
```

```
After balancing: [639 639]
```

5. Model Training

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
```

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Code JupyterLab Python 3 (pykernel)

5. Model Training

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

# dictionary of classifiers
models = {
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(random_state=42),
    "XGBoost": XGBClassifier(random_state=42)
}
```

```
# dictionary to store the cross validation results
cv_scores = {}

# perform 5-fold cross validation for each model
for model_name, model in models.items():
  print(f"Training {model_name} with default parameters...")
  scores = cross_val_score(model, X_train_smote, y_train_smote, cv=5, scoring="accuracy")
  cv_scores[model_name] = scores
  print(f"{model_name} Cross-Validation Accuracy: {np.mean(scores):.2f}")
  print("-"*50)
```

```
Training Decision Tree with default parameters...
Decision Tree Cross-Validation Accuracy: 0.86
--------------------------------------------------
Training Random Forest with default parameters...
Random Forest Cross-Validation Accuracy: 0.92
--------------------------------------------------
Training XGBoost with default parameters...
```

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Code JupyterLab Python 3 (pykernel)

```
Training Decision Tree with default parameters...
Decision Tree Cross-Validation Accuracy: 0.86
--------------------------------------------------
Training Random Forest with default parameters...
Random Forest Cross-Validation Accuracy: 0.92
--------------------------------------------------
Training XGBoost with default parameters...
XGBoost Cross-Validation Accuracy: 0.90
--------------------------------------------------
```

[ ]:
```
cv_scores
```

[66]:
```
{'Decision Tree': array([0.7961165 , 0.87864078, 0.87378641, 0.8592233 , 0.87378641]),
 'Random Forest': array([0.90776699, 0.92718447, 0.9223301 , 0.91747573, 0.9223301 ]),
 'XGBoost': array([0.87378641, 0.9223301 , 0.89320388, 0.91262136, 0.91747573])}
```

6. Model Selection & Hyperparameter Tuning

[ ]:
```
# Initializing models
decision_tree = DecisionTreeClassifier(random_state=42)
random_forest = RandomForestClassifier(random_state=42)
xgboost_classifier = XGBClassifier(random_state=42)
```

[ ]:
```
# Hyperparameter grids for RandomizedSearchCV

param_grid_dt = {
    "criterion": ["gini", "entropy"],
    "max_depth": [None, 10, 20, 30, 50, 70],
    "min_samples_split": [2, 5, 10],
    "min_samples_leaf": [1, 2, 4]
}
```

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Code | JupyterLab | Python 3 (pykernel)

6. Model Selection & Hyperparameter Tuning

```
# Initializing models
decision_tree = DecisionTreeClassifier(random_state=42)
random_forest = RandomForestClassifier(random_state=42)
xgboost_classifier = XGBClassifier(random_state=42)
```

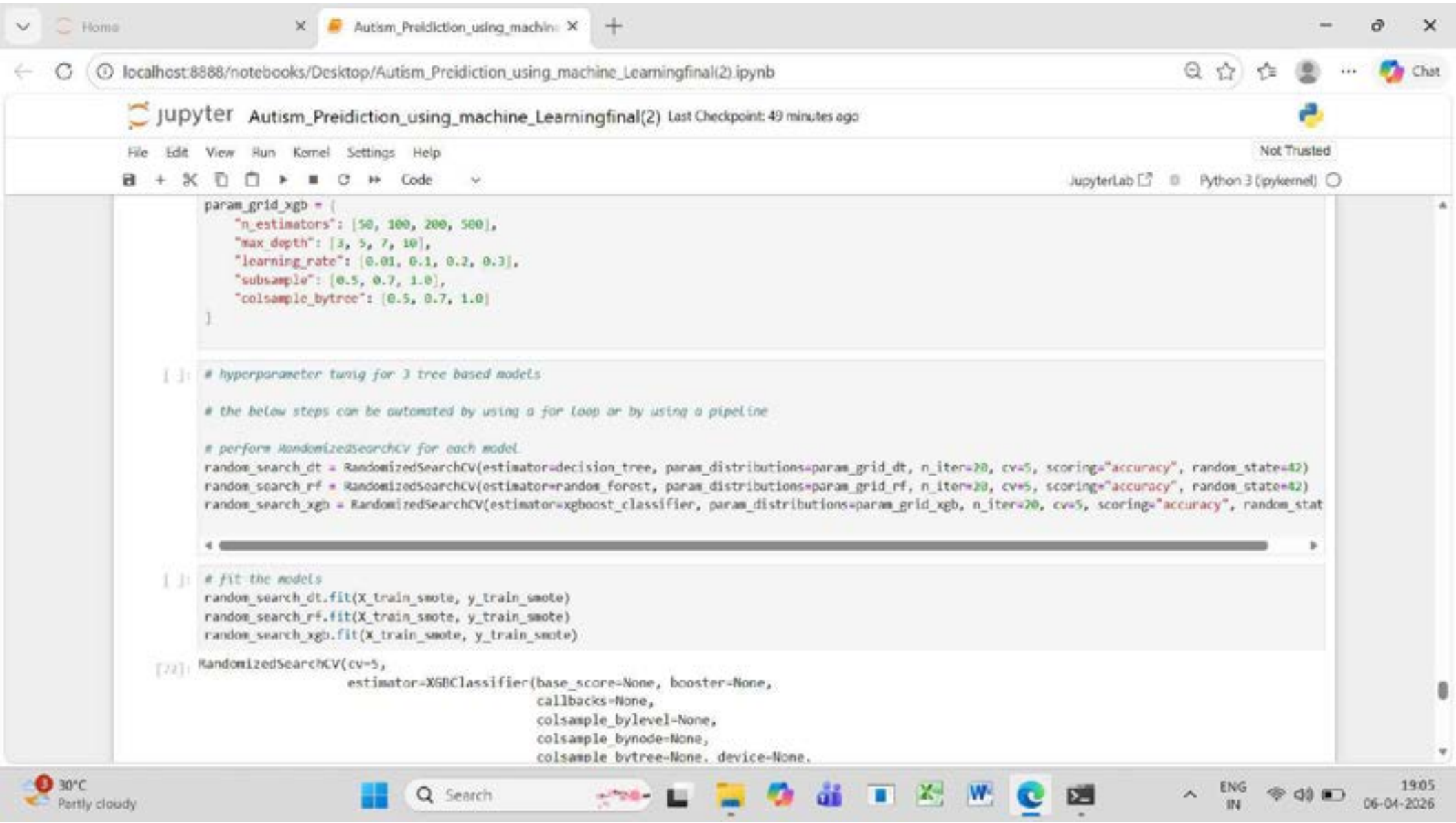
```
# Hyperparameter grids for RandomizedSearchCV

param_grid_dt = {
    "criterion": ["gini", "entropy"],
    "max_depth": [None, 10, 20, 30, 50, 70],
    "min_samples_split": [2, 5, 10],
    "min_samples_leaf": [1, 2, 4]
}


param_grid_rf = {
    "n_estimators": [50, 100, 200, 500],
    "max_depth": [None, 10, 20, 30],
    "min_samples_split": [2, 5, 10],
    "min_samples_leaf": [1, 2, 4],
    "bootstrap": [True, False]
}


param_grid_xgb = {
    "n_estimators": [50, 100, 200, 500],
    "max_depth": [3, 5, 7, 10],
```

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Code
JupyterLab
Python 3 (pykernel)
param_grid_xgb = {
"n_estimators": [50, 100, 200, 500],
"max_depth": [3, 5, 7, 10],
"learning_rate": [0.01, 0.1, 0.2, 0.3],
"subsample": [0.5, 0.7, 1.0],
"colsample_bytree": [0.5, 0.7, 1.0]
}
hyperparameter tuning for 3 tree based models
the below steps can be automated by using a for loop or by using a pipeline
perform RandomizedSearchCV for each model
random_search_dt = RandomizedSearchCV(estimator=decision_tree, param_distributions=param_grid_dt, n_iter=20, cv=5, scoring="accuracy", random_state=42)
random_search_rf = RandomizedSearchCV(estimator=random_forest, param_distributions=param_grid_rf, n_iter=20, cv=5, scoring="accuracy", random_state=42)
random_search_xgb = RandomizedSearchCV(estimator=xgboost_classifier, param_distributions=param_grid_xgb, n_iter=20, cv=5, scoring="accuracy", random_stat
fit the models
random_search_dt.fit(X_train_smote, y_train_smote)
random_search_rf.fit(X_train_smote, y_train_smote)
random_search_xgb.fit(X_train_smote, y_train_smote)
RandomizedSearchCV(cv=5,
estimator=XGBClassifier(base_score=None, booster=None,
callbacks=None,
colsample_bylevel=None,
colsample_bynode=None,
colsample_bytree=None, device=None,
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Code

JupyterLab | Python 3 (pykernel)

[ ]:
```
# fit the models
random_search_dt.fit(X_train_smote, y_train_smote)
random_search_rf.fit(X_train_smote, y_train_smote)
random_search_xgb.fit(X_train_smote, y_train_smote)
```

[72]:
```
RandomizedSearchCV(cv=5,
                   estimator=XGBClassifier(base_score=None, booster=None,
                                           callbacks=None,
                                           colsample_bylevel=None,
                                           colsample_bynode=None,
                                           colsample_bytree=None, device=None,
                                           early_stopping_rounds=None,
                                           enable_categorical=False,
                                           eval_metric=None, feature_types=None,
                                           feature_weights=None, gamma=None,
                                           grow_policy=None,
                                           importance_type=None,
                                           interaction_constrain...
                                           min_child_weight=None, missing=nan,
                                           monotone_constraints=None,
                                           multi_strategy=None,
                                           n_estimators=None, n_jobs=None,
                                           num_parallel_tree=None, ...),
                   n_iter=20,
                   param_distributions={'colsample_bytree': [0.5, 0.7, 1.0],
                                        'learning_rate': [0.01, 0.1, 0.2, 0.3],
                                        'max_depth': [3, 5, 7, 10],
                                        'n_estimators': [50, 100, 200, 500],
                                        'subsample': [0.5, 0.7, 1.0]},
                   random_state=42, scoring='accuracy')
```

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Code JupyterLab Python 3 (pykernel)

```
                    n_iter=20,
                    param_distributions={'colsample_bytree': [0.5, 0.7, 1.0],
                                         'learning_rate': [0.01, 0.1, 0.2, 0.3],
                                         'max_depth': [3, 5, 7, 10],
                                         'n_estimators': [50, 100, 200, 500],
                                         'subsample': [0.5, 0.7, 1.0]},
                    random_state=42, scoring='accuracy')
```

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[ ]:

```
# Get the model with best score

best_model = None
best_score = 0

# Check if random_search_dt was successfully executed
if 'random_search_dt' in locals() and random_search_dt.best_score_ > best_score:
  best_model = random_search_dt.best_estimator_
  best_score = random_search_dt.best_score_

# Check if random_search_rf was successfully executed
if 'random_search_rf' in locals() and random_search_rf.best_score_ > best_score:
  best_model = random_search_rf.best_estimator_
  best_score = random_search_rf.best_score_

# Check if random_search_xgb was successfully executed
if 'random_search_xgb' in locals() and random_search_xgb.best_score_ > best_score:
  best_model = random_search_xgb.best_estimator_
  best_score = random_search_xgb.best_score_
```

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Code JupyterLab Python 3 (pykernel)

```
[ ]: # Get the model with best score

best_model = None
best_score = 0

# Check if random_search_dt was successfully executed
if 'random_search_dt' in locals() and random_search_dt.best_score_ > best_score:
  best_model = random_search_dt.best_estimator_
  best_score = random_search_dt.best_score_

# Check if random_search_rf was successfully executed
if 'random_search_rf' in locals() and random_search_rf.best_score_ > best_score:
  best_model = random_search_rf.best_estimator_
  best_score = random_search_rf.best_score_

# Check if random_search_xgb was successfully executed
if 'random_search_xgb' in locals() and random_search_xgb.best_score_ > best_score:
  best_model = random_search_xgb.best_estimator_
  best_score = random_search_xgb.best_score_
```

```
[ ]: # print information about the voting model since it's used for evaluation
print(f"Voting Model: {voting_model}")

# Assuming you have already calculated the accuracy score for the voting model
# If not, you would need to calculate it first:
# from sklearn.metrics import accuracy_score
# y_pred_voting = voting_model.predict(X_test)
# voting_accuracy = accuracy_score(y_test, y_pred_voting)
# print(f"Voting Model Accuracy: {voting_accuracy:.2f}")
```

Voting Model: VotingClassifier(estimators=[('xgb'

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Code

JupyterLab | Python 3 (pykernel)

[ ]:
```
# print information about the voting model since it's used for evaluation
print(f"Voting Model: {voting_model}")

# Assuming you have already calculated the accuracy score for the voting model
# If not, you would need to calculate it first:
# from sklearn.metrics import accuracy_score
# y_pred_voting = voting_model.predict(X_test)
# voting_accuracy = accuracy_score(y_test, y_pred_voting)
# print(f"Voting Model Accuracy: {voting_accuracy:.2f}")
```

```
Voting Model: VotingClassifier(estimators=[('xgb',
                               XGBClassifier(base_score=None, booster=None,
                                             callbacks=None,
                                             colsample_bylevel=None,
                                             colsample_bynode=None,
                                             colsample_bytree=0.9, device=None,
                                             early_stopping_rounds=None,
                                             enable_categorical=False,
                                             eval_metric='logloss',
                                             feature_types=None,
                                             feature_weights=None, gamma=None,
                                             grow_policy=None,
                                             importance_type=None,
                                             interaction_con...
                                             max_cat_threshold=None,
                                             max_cat_to_onehot=None,
                                             max_delta_step=None, max_depth=6,
                                             max_leaves=None,
                                             min_child_weight=None, missing=nan,
                                             monotone_constraints=None,
                                             multi_strategy=None,
                                             n_estimators=400, n_jobs=None,
```

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Code JupyterLab Python 3 (pykernel)

```
Voting Model: VotingClassifier(estimators=[('xgb',
                                XGBClassifier(base_score=None, booster=None,
                                              callbacks=None,
                                              colsample_bylevel=None,
                                              colsample_bynode=None,
                                              colsample_bytree=0.9, device=None,
                                              early_stopping_rounds=None,
                                              enable_categorical=False,
                                              eval_metric='logloss',
                                              feature_types=None,
                                              feature_weights=None, gamma=None,
                                              grow_policy=None,
                                              importance_type=None,
                                              interaction_con...
                                              max_cat_threshold=None,
                                              max_cat_to_onehot=None,
                                              max_delta_step=None, max_depth=6,
                                              max_leaves=None,
                                              min_child_weight=None, missing=nan,
                                              monotone_constraints=None,
                                              multi_strategy=None,
                                              n_estimators=400, n_jobs=None,
                                              num_parallel_tree=None, ...)),
                               ('rf',
                                RandomForestClassifier(n_estimators=300,
                                                       random_state=42)),
                               ('lr',
                                LogisticRegression(max_iter=1000,
                                                   random_state=42))],
                   voting='soft')
```

```
[ ]: # save the best model
```

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Code JupyterLab Python 3 (pykernel)

```
[ ]: # save the best model
with open("best_model.pkl", "wb") as f:
    pickle.dump(best_model, f)
```

```
[ ]: from sklearn.ensemble import VotingClassifier, RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from xgboost import XGBClassifier

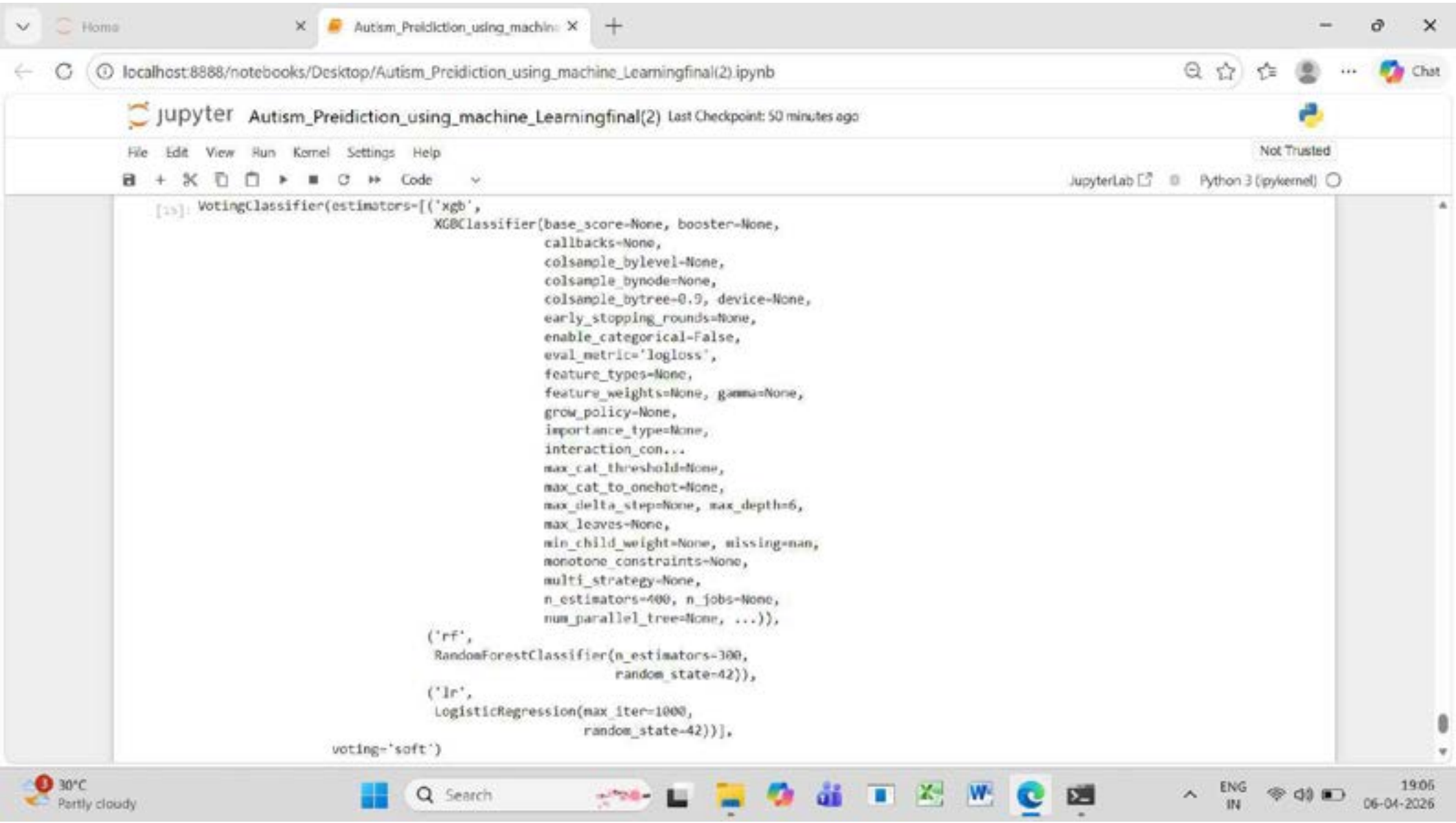
xgb = XGBClassifier(
    n_estimators=400,
    learning_rate=0.05,
    max_depth=6,
    subsample=0.9,
    colsample_bytree=0.9,
    random_state=42,
    eval_metric='logloss'
)
rf = RandomForestClassifier(n_estimators=300, random_state=42)
lr = LogisticRegression(max_iter=1000, random_state=42)

voting_model = VotingClassifier(
    estimators=[('xgb', xgb), ('rf', rf), ('lr', lr)],
    voting='soft'
)

voting_model.fit(X_res, y_res)
```

```
[15]: VotingClassifier(estimators=[('xgb',
                                   XGBClassifier(base_score=None, booster=None,
```

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Python 3 (pykernel)
[19]:
VotingClassifier(estimators=[('xgb',
XGBClassifier(base_score=None, booster=None,
callbacks=None,
colsample_bylevel=None,
colsample_bynode=None,
colsample_bytree=0.9, device=None,
early_stopping_rounds=None,
enable_categorical=False,
eval_metric='logloss',
feature_types=None,
feature_weights=None, gamma=None,
grow_policy=None,
importance_type=None,
interaction_con...
max_cat_threshold=None,
max_cat_to_onehot=None,
max_delta_step=None, max_depth=6,
max_leaves=None,
min_child_weight=None, missing=nan,
monotone_constraints=None,
multi_strategy=None,
n_estimators=400, n_jobs=None,
num_parallel_tree=None, ...)),
('rf',
RandomForestClassifier(n_estimators=300,
random_state=42)),
('lr',
LogisticRegression(max_iter=1000,
random_state=42))],
voting='soft')
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```
                                RandomForestClassifier(n_estimators=300,
                                                       random_state=42)),
                               ('lr',
                                LogisticRegression(max_iter=1000,
                                                   random_state=42))],
                   voting='soft')
```

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7. Evaluation

```
# Evaluate the voting model
y_pred = voting_model.predict(X_test) # Changed best_model to voting_model

print(f"Final Accuracy: {accuracy_score(y_test, y_pred)*100:.2f}%")
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Final Accuracy: 98.12%
[[122   2]
 [  1  35]]
              precision    recall  f1-score   support

           0       0.99      0.98      0.99       124
           1       0.95      0.97      0.96        36

    accuracy                           0.98       160
   macro avg       0.97      0.98      0.97       160
weighted avg       0.98      0.98      0.98       160
```

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Code | JupyterLab | Python 3 (ipykernel)

[1]:
```
import matplotlib.pyplot as plt
import numpy as np
# Step 1: Define metrics
labels = ["Accuracy", "Precision", "Recall", "F1-Score"]
# Step 2: Enter your model values (REPLACE with your actual results)
log_reg = [88, 85, 86, 85]
rf = [92, 90, 91, 90]
xgb = [96, 95, 94, 95]
ensemble = [98, 97, 97, 97]

# Step 3: Prepare angle values
angles = np.linspace(0, 2*np.pi, len(labels), endpoint=False).tolist()
angles += angles[:1]  # close the circle

# Step 4: Function to plot
def plot_model(values, label):
    values = values + values[:1]  # close shape
    plt.plot(angles, values, label=label)
    plt.fill(angles, values, alpha=0.1)

# Step 5: Create radar chart
plt.figure(figsize=(6,6))
ax = plt.subplot(polar=True)

# Step 6: Plot all models
plot_model(log_reg, "Logistic Regression")
plot_model(rf, "Random Forest")
plot_model(xgb, "XGBoost")
plot_model(ensemble, "Ensemble Voting")

# Step 7: Labels & title
plt.xticks(angles[:-1], labels)
plt.title("Model Performance Comparison (Radar Chart)")
plt.legend(loc='upper right', bbox_to_anchor=(1.3, 1.1))

# Step 8: Show graph
plt.show()
```

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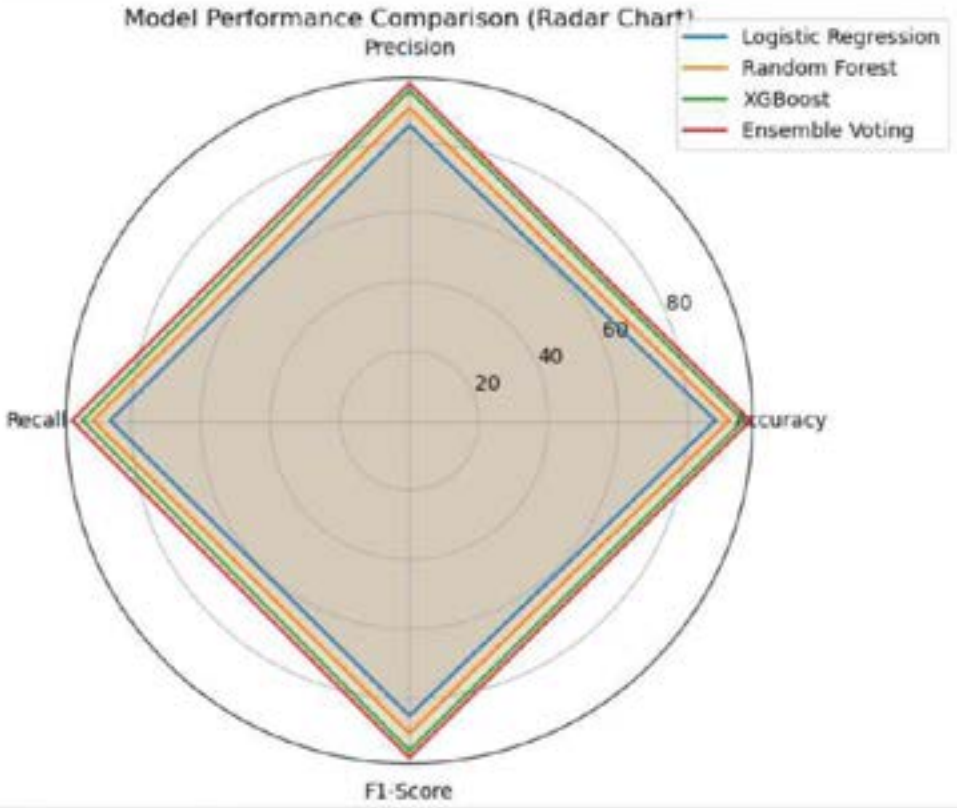
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Code

JupyterLab

Python 3 (pykernel)



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